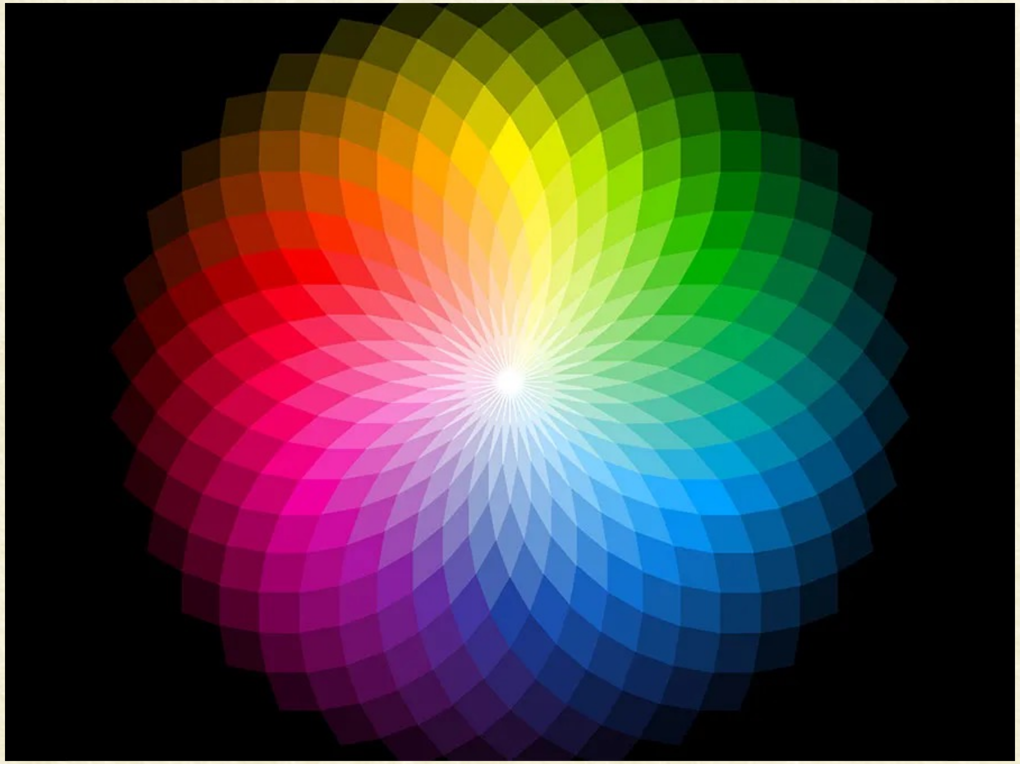
CS7.404: Digital Image Processing

Monsoon 2025: Color Image Processing



Anoop M. Namboodiri

Biometrics and Secure ID Lab, CVIT,

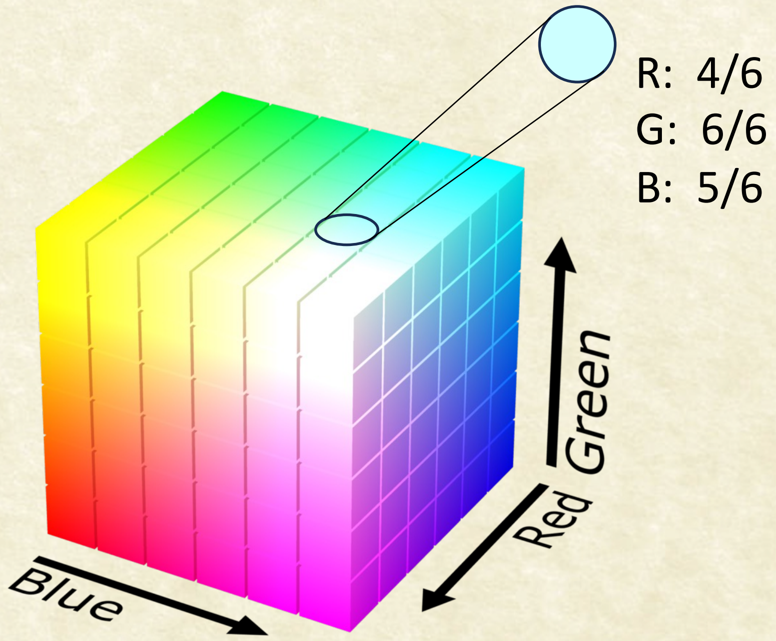
IIIT Hyderabad

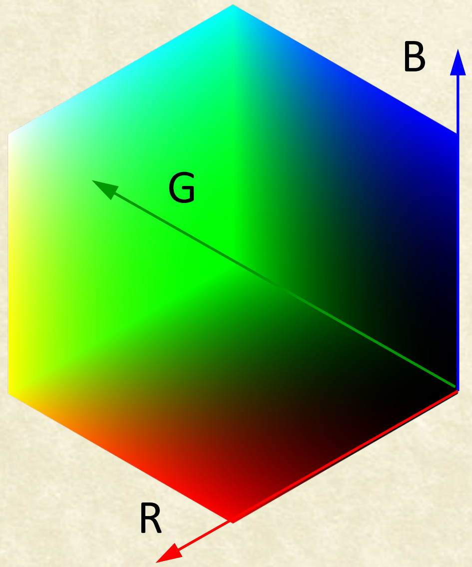
RGB color space

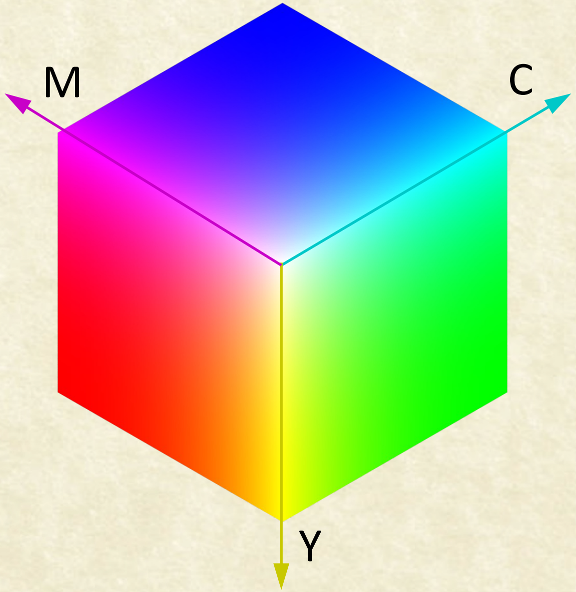
- Primary colors
- Additive color model

$$f(x, y) = \alpha_1 R + \alpha_2 G + \alpha_3 B$$

- Challenge: Perceptually non uniform
- RGB vs CMY

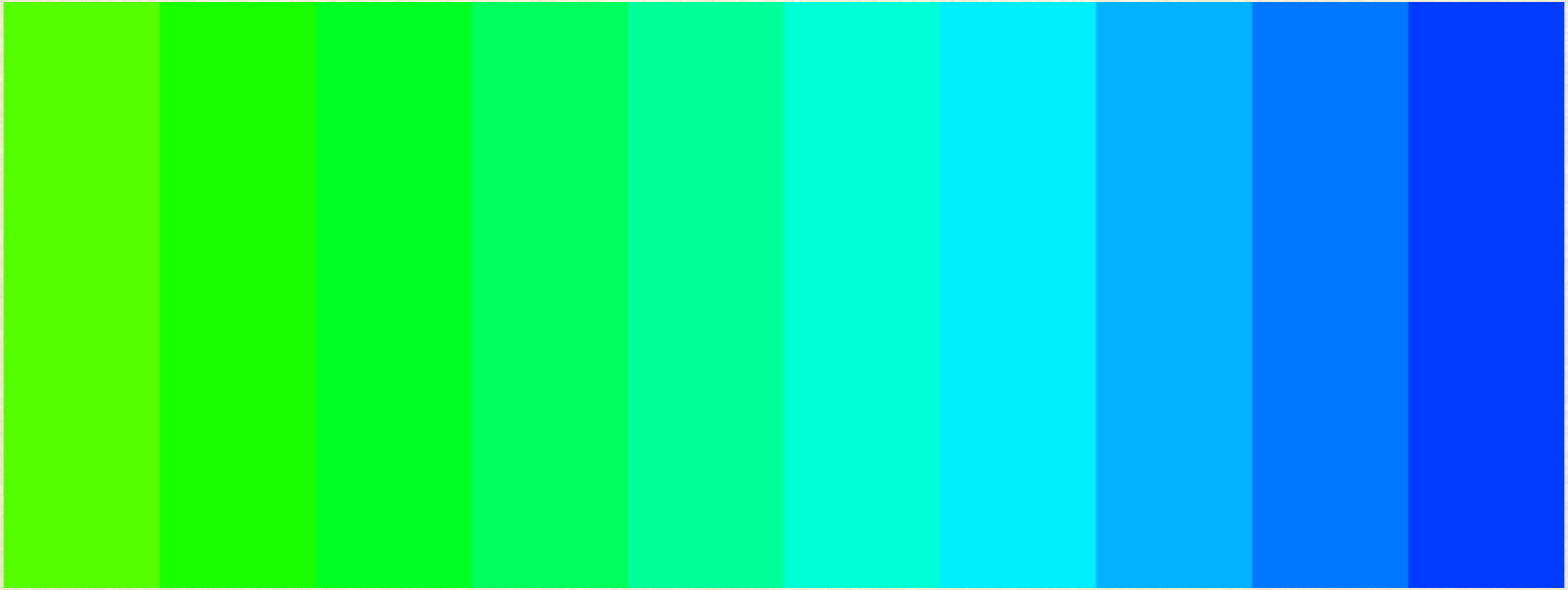






RGB : Non-Uniform Perceptual Space

- A given numerical change does not result in similar perceived change in color.



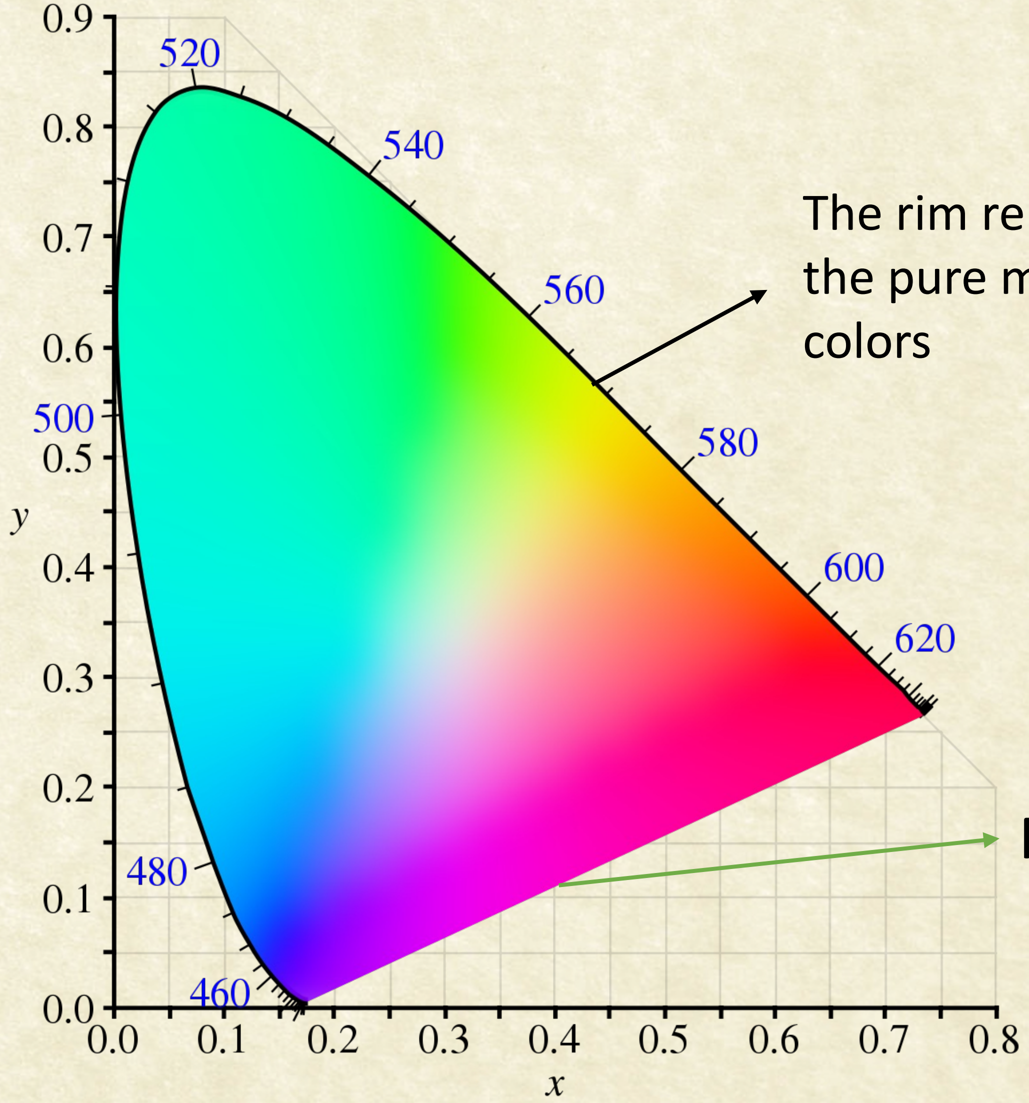
CIE 1931 (x,y) chromaticity diagram

- Created by the Intl. Commission on Illumination (CIE) in 1931
- XYZ: Axes beyond human perception
- Y x y: Luminance + the two most distinctive chrominance components:

$$x = \frac{X}{X+Y+Z}, \qquad y = \frac{Y}{X+Y+Z}$$

- Separate luminance and chromaticity

$$\begin{bmatrix} R \\ G \\ B \end{bmatrix} = \begin{bmatrix} 3.063 & -1.393 & -0.476 \\ -0.969 & 1.876 & 0.042 \\ 0.068 & -0.229 & 1.069 \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \end{bmatrix}$$

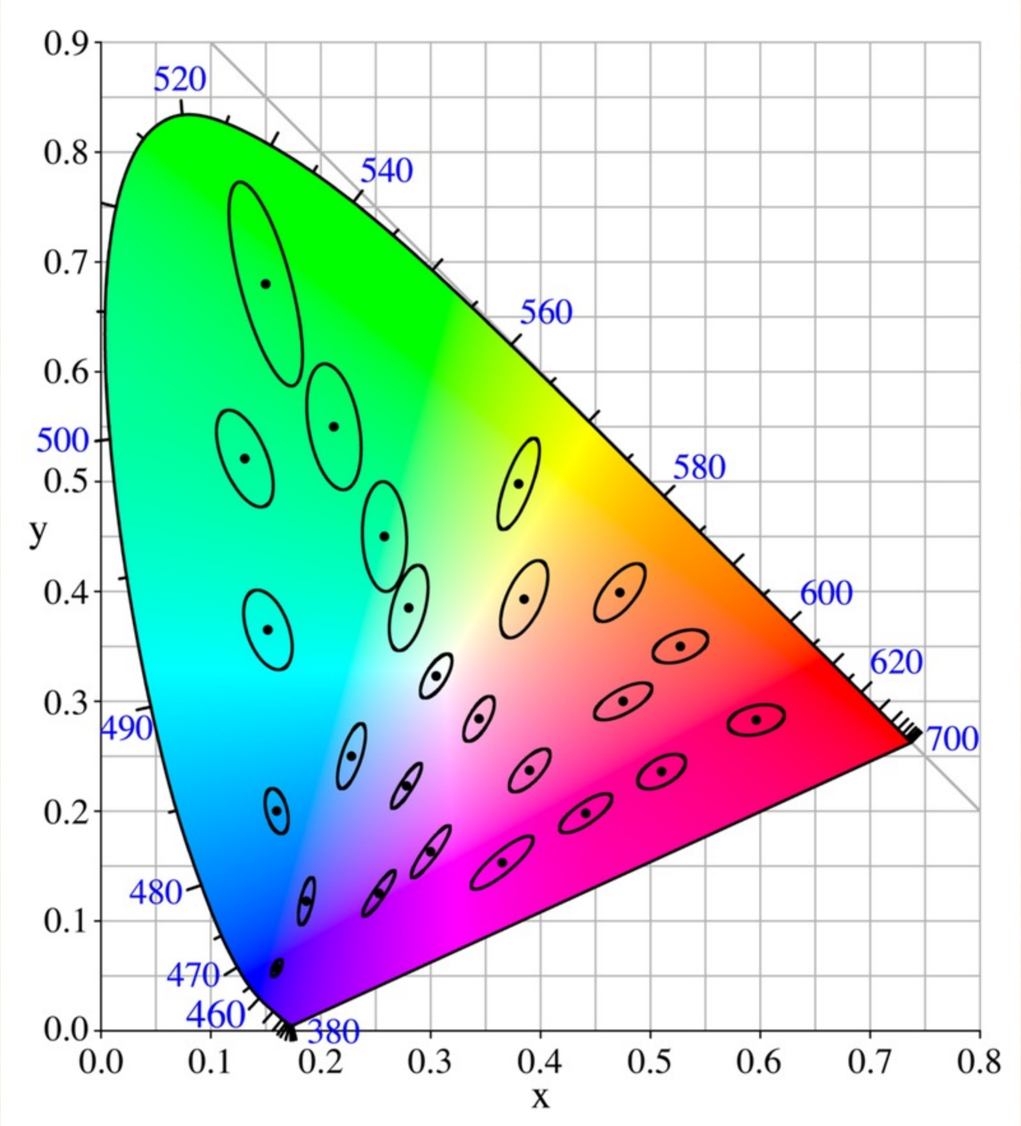


The rim represents all of the pure monochromatic colors

Purple Line

Visualizing CIE XYZ color space: https://www.youtube.com/watch?v=x0-qoXOCOow

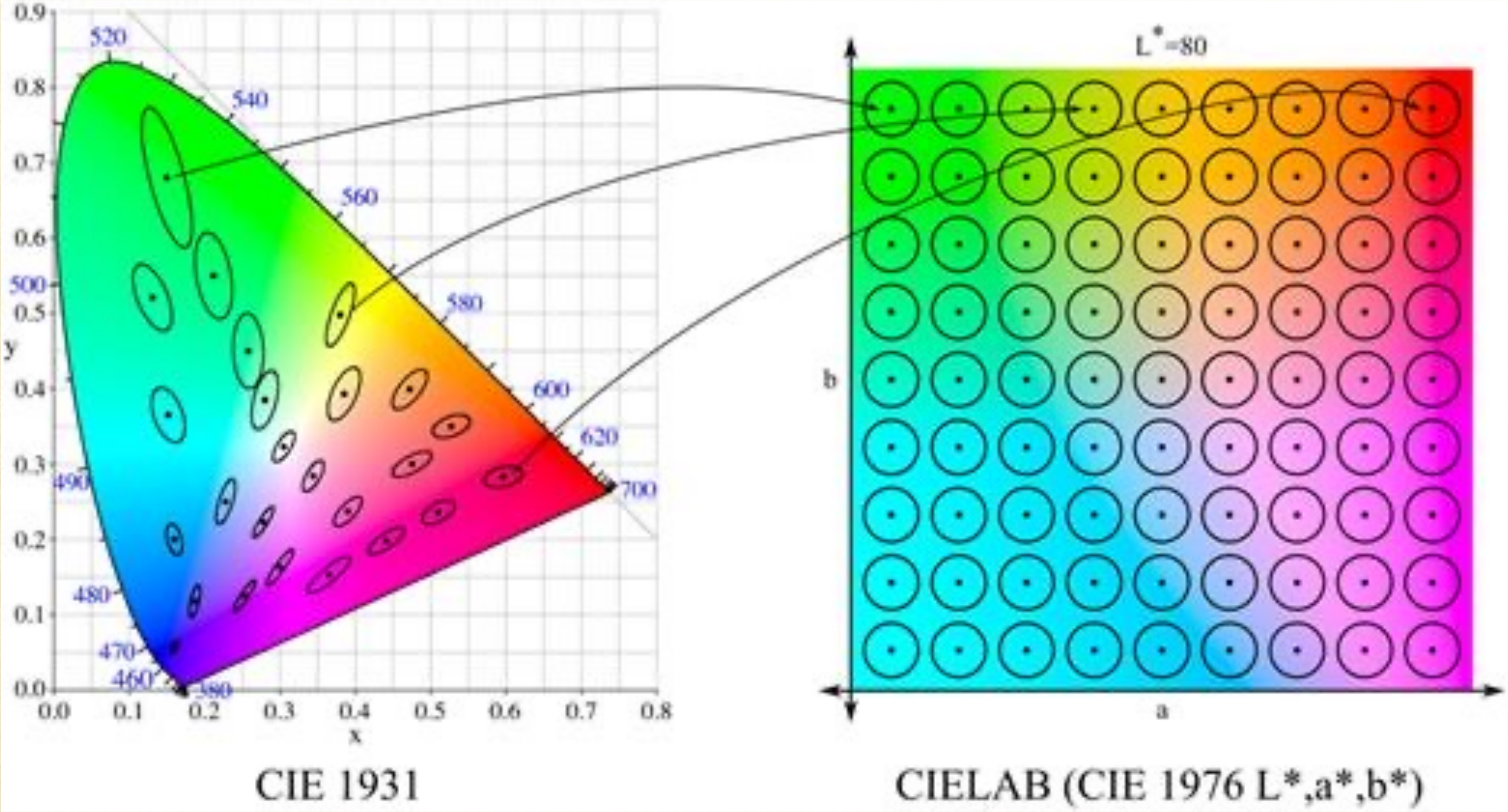
McAdam Ellipses



- **McAdam ellipses:** Regions on a chromaticity diagram, where colors are indistinguishable from the center of the ellipse, to the average human eye.
- The contour of the ellipse represents the just noticeable differences of chromaticity

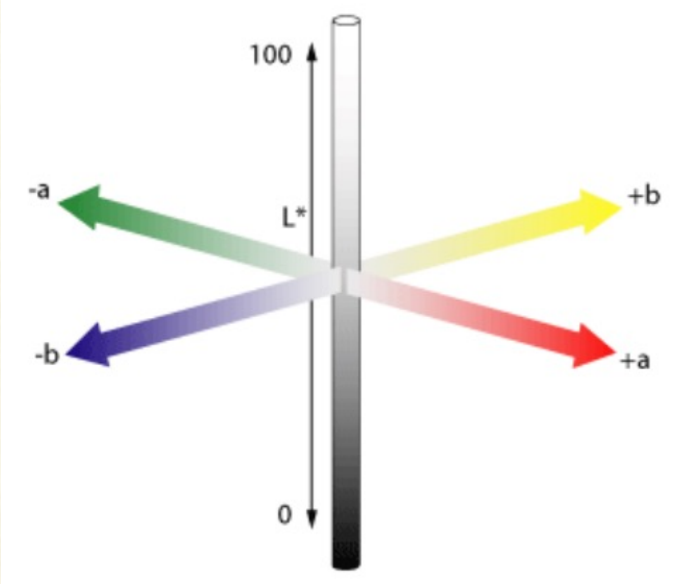
courtesy: wikipedia

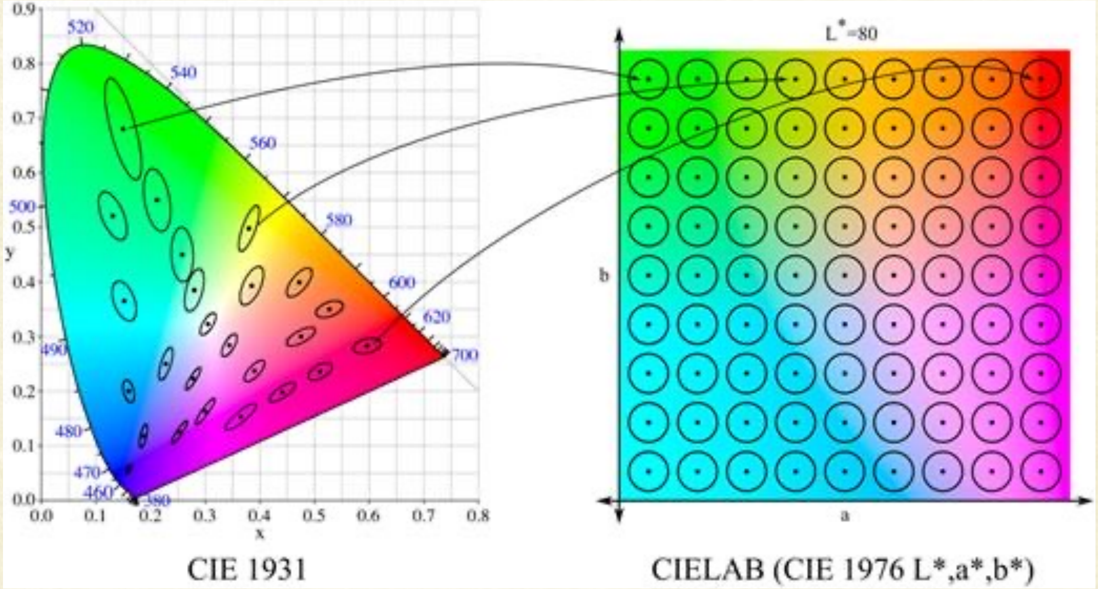
CIE Lab color space



Ideal scenario

CIE Lab color space



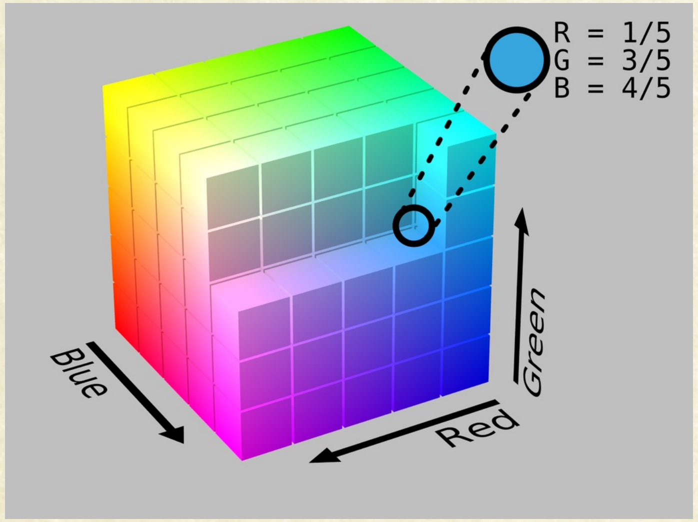


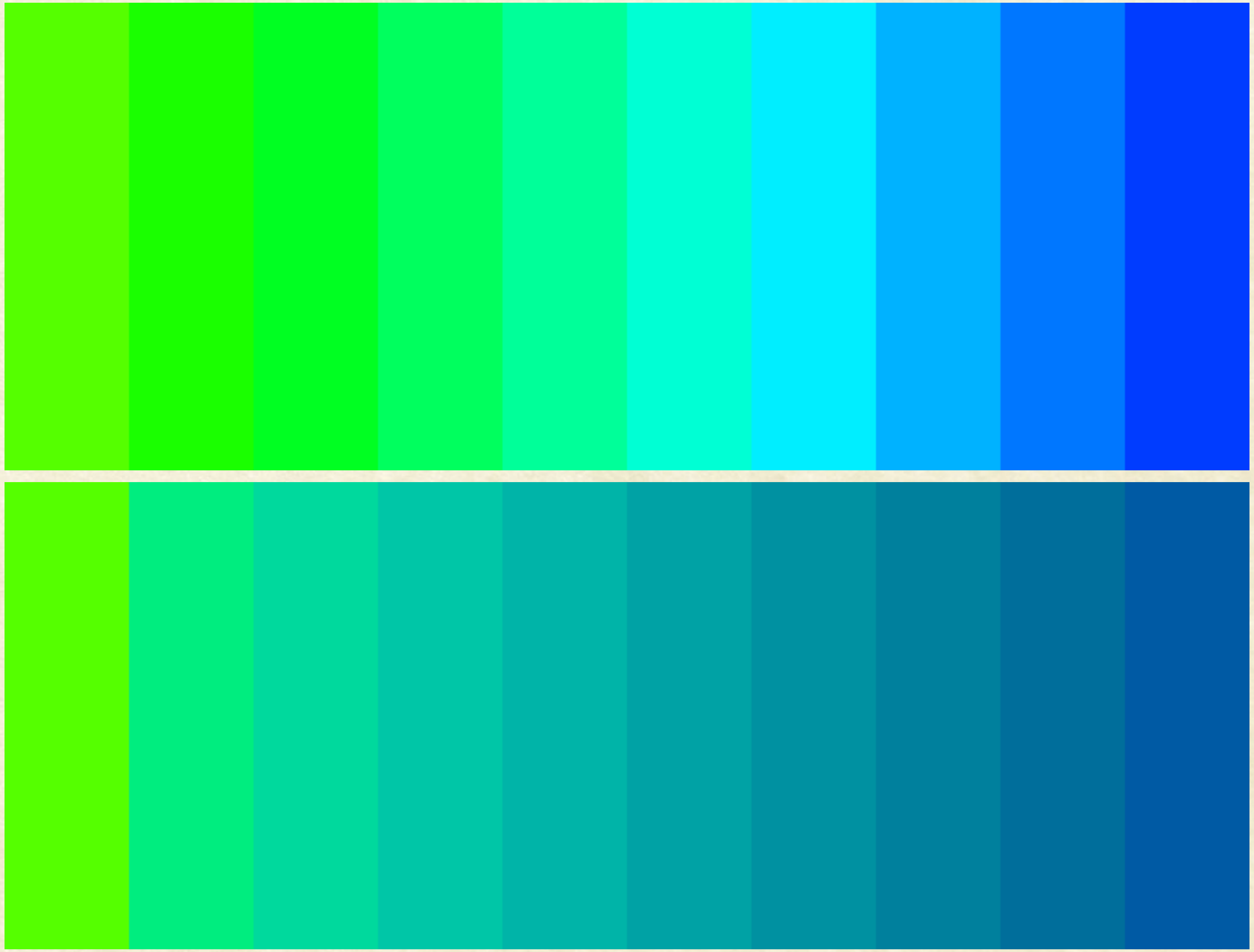
Components:

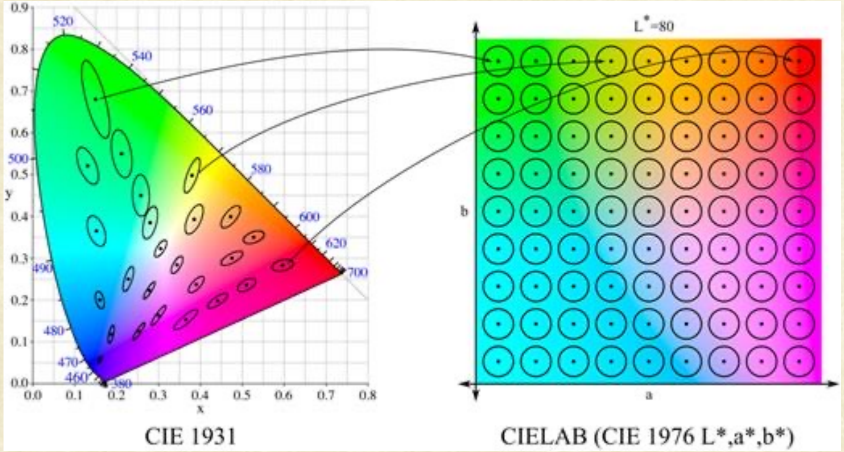
- Luminosity L
- Color components
  - a* : color hue, saturation along green-red axis
  - b* : color hue, saturation along blue-yellow axis

- Makes changes in CIE space linear with respect to human perception
- Popular in high-quality photographic applications

Perceptually Uniform Color Spaces







https://matplotlib.org/3.1.1/tutorials/colors/colormaps.html

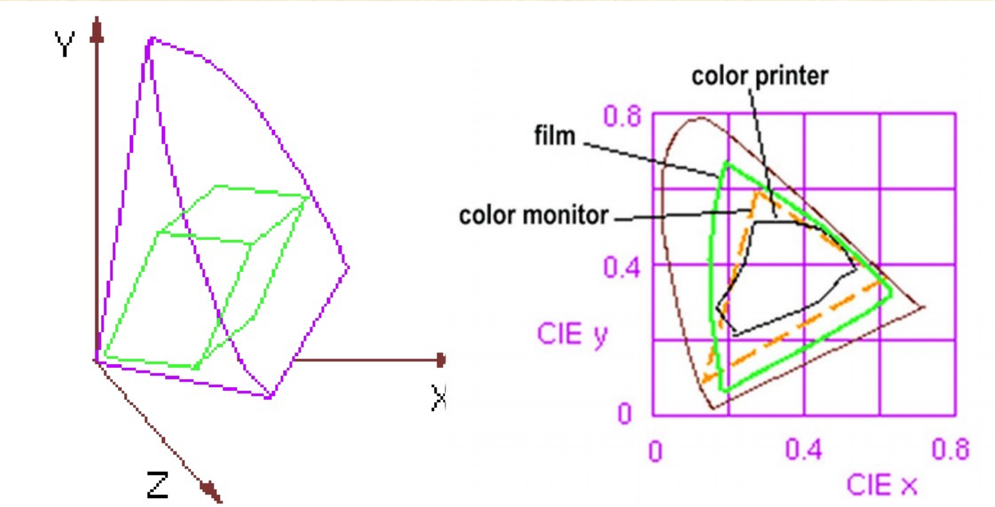
Measuring Color Differences

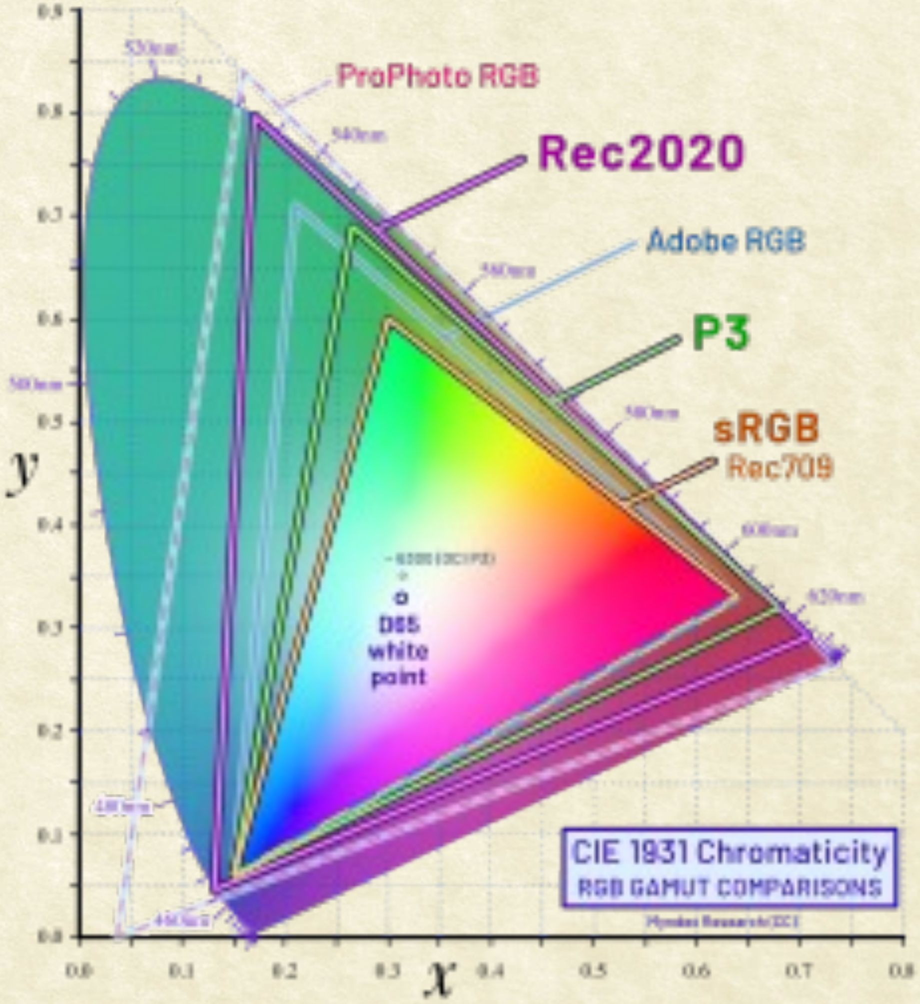
- Due to its uniformity with respect to human perception, the differences between colors in L*a*b* color space can be determined as **Euclidean distance**.

$$
\begin{aligned}
\mathrm{ColorDist}_{\mathrm{Lab}}(\mathbf{C}_1, \mathbf{C}_2) &= \|\mathbf{C}_1 - \mathbf{C}_2\| \\
&= \sqrt{(L_1^* - L_2^*)^2 + (a_1^* - a_2^*)^2 + (b_1^* - b_2^*)^2}
\end{aligned}
$$

Color Gamut of Devices

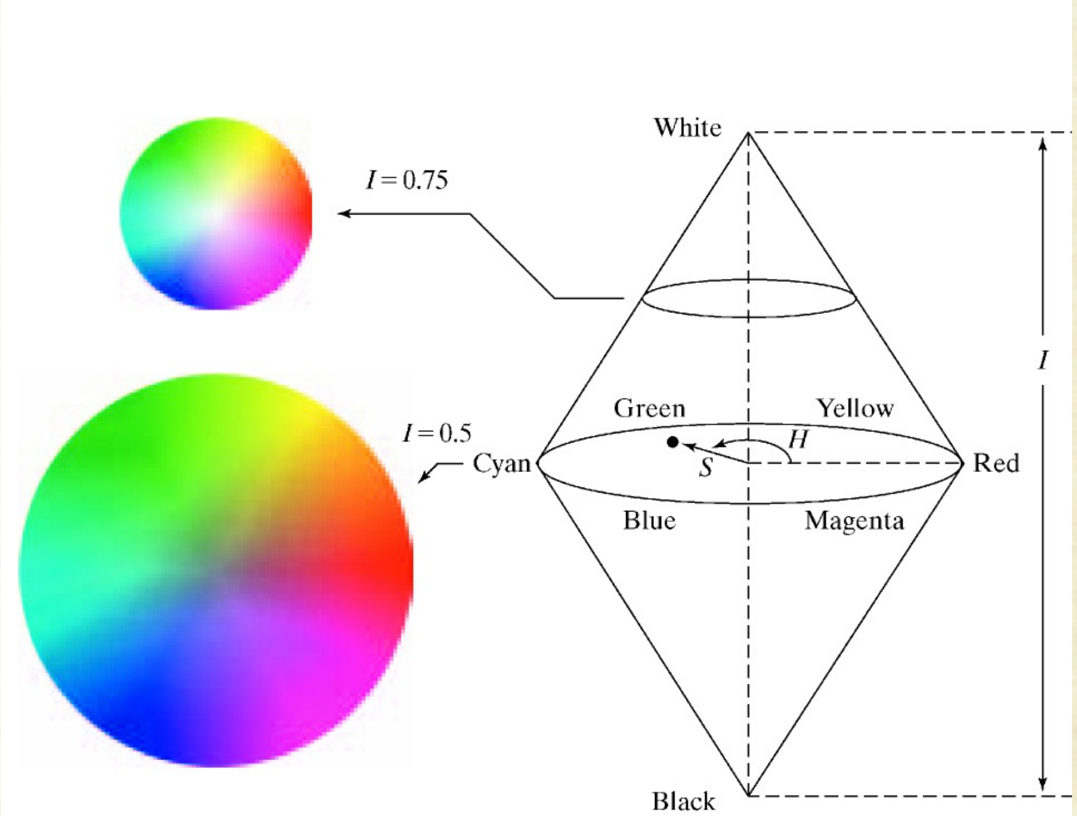
- The RGB color cube sits within the CIE color space
- CIE chromaticity diagram is useful to compare the gamut of various devices
  - e.g., compare color printer and monitor color gamut.





HSI color space

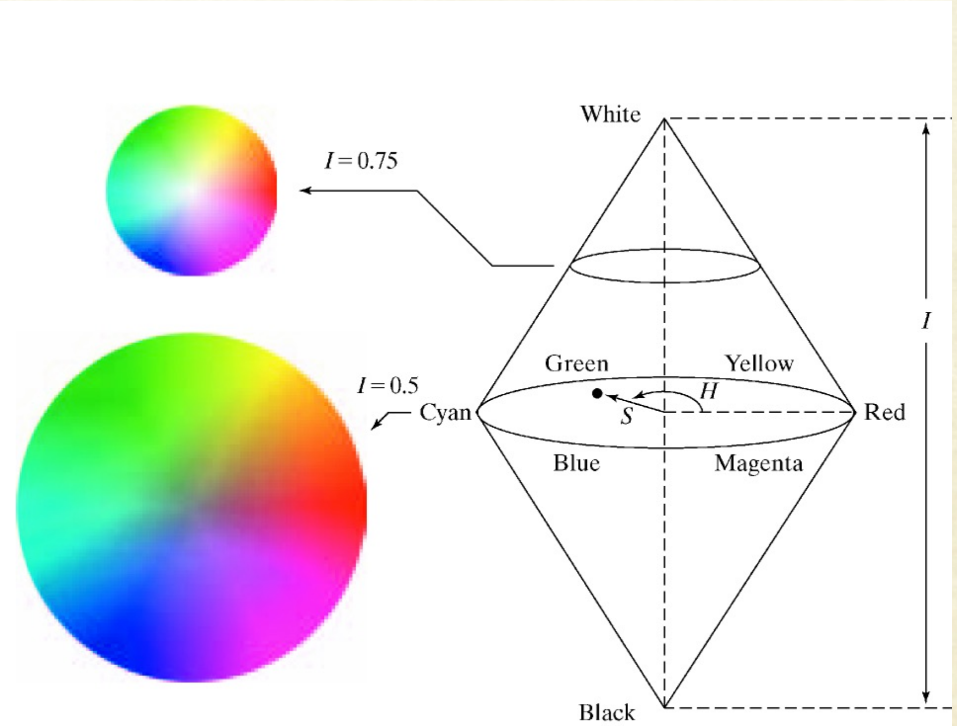
- Hue (H):
  - Dominant colour perceived by an observer
  - When we call an object red or orange, we refer to its hue.
- Saturation (S):
  - Amount of white light mixed with a hue
  - Pure colors are fully saturated.
    - Pink (red+white) is less saturated.
- Brightness (I) : achromatic notion of intensity



HSI color space

- **HUE**
  - A subjective measure of color
  - Average human eye can perceive ~200 different colors
- **Saturation**
  - Relative purity of the color. Mixing more "white" with color reduces its saturation.
  - Pink has the same **hue** as **red** but less **saturation**
- **Intensity**
  - The brightness or darkness of an object

RGB : great for color generation
HSI : great for color description

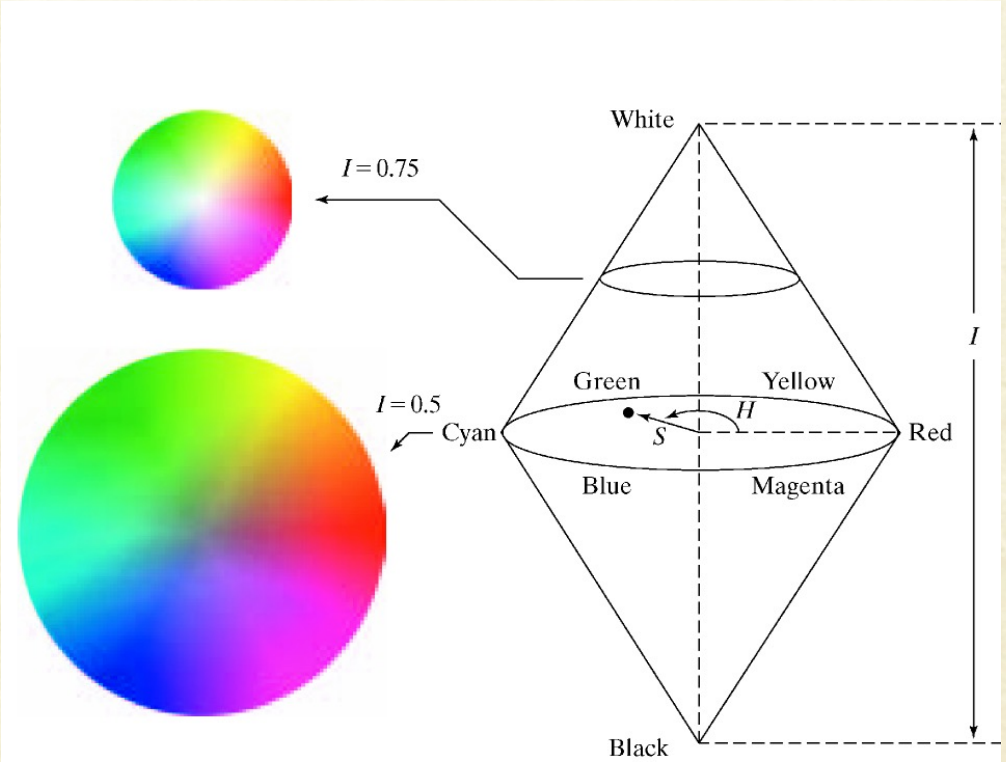


HSI color space

- **Hue** is defined as an angle
  - 0 degrees is **RED**
  - 120 degrees is **GREEN**
  - 240 degrees is **BLUE**

Saturation = % of distance from center : [0,1]

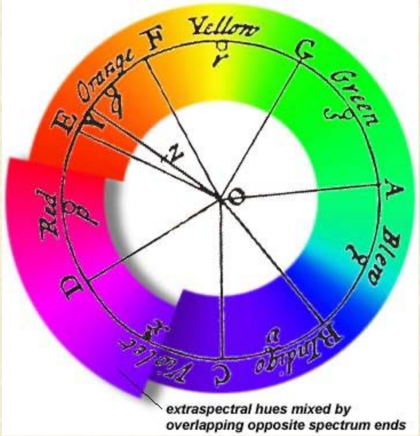
- **Intensity** is denoted as the distance "up" the axis from black.
  - Values range from 0 to 1



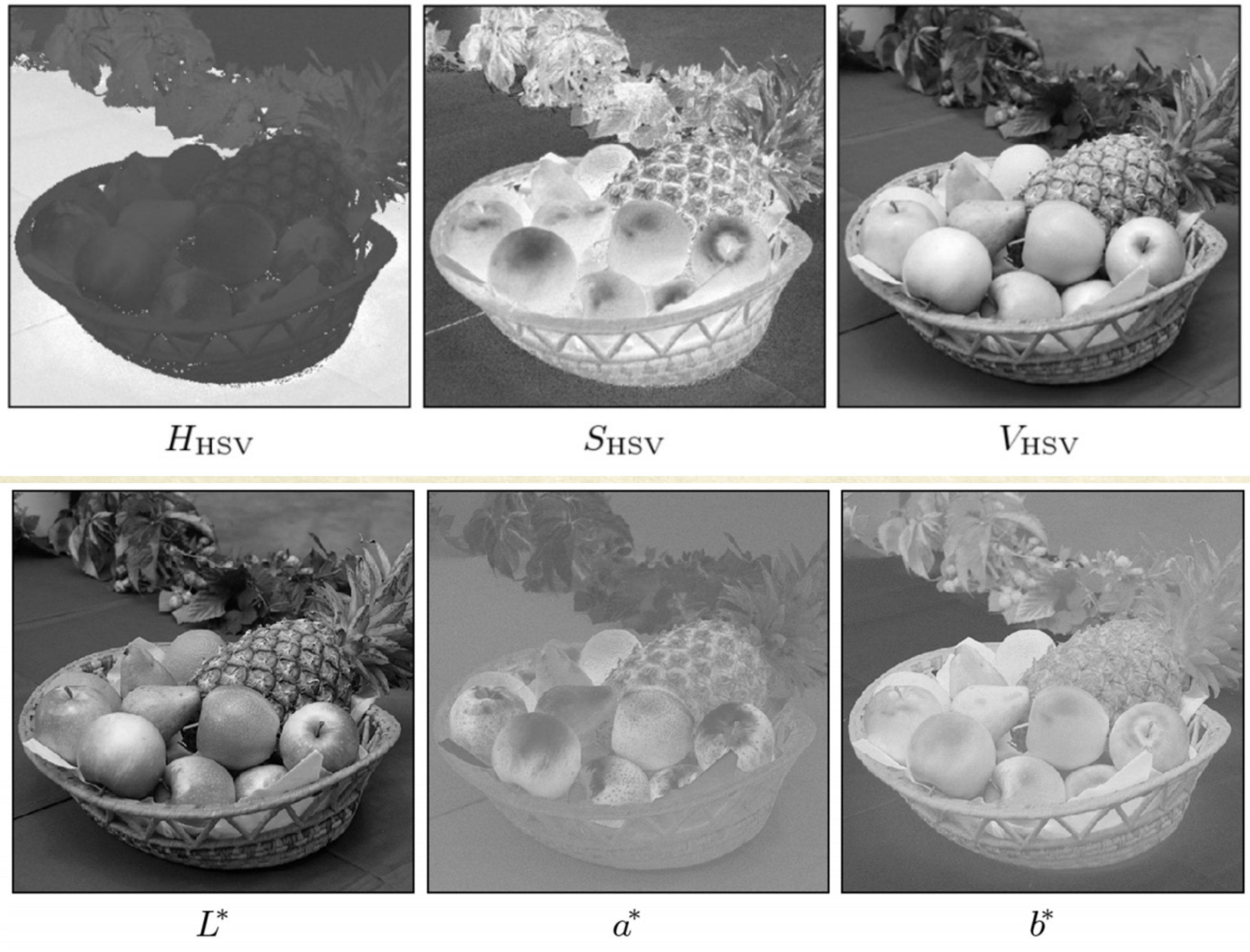
- RGB to HSI transformation:

Define: $\alpha = R - \frac{1}{2}(G + B); and\ \beta = \frac{\sqrt{3}}{2}(G - B)$

$$\begin{cases} H = \arctan(\beta/\alpha) \\ S = \sqrt{\alpha^2 + \beta^2} \\ I = (R + G + B)/3 \end{cases}$$

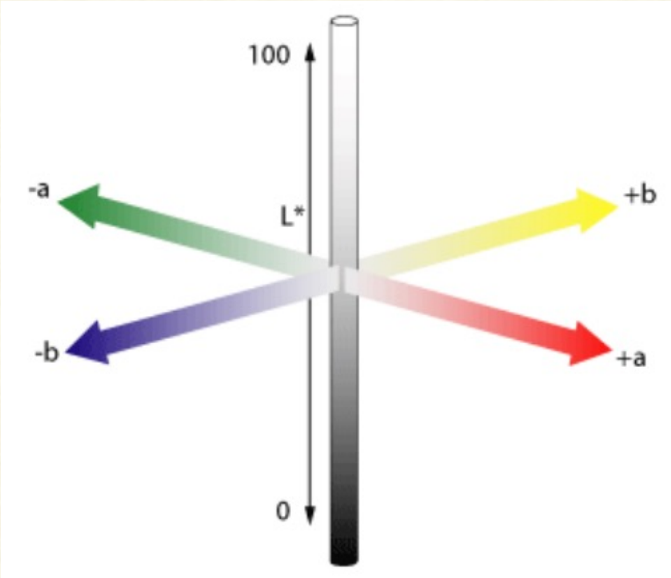


HSI vs Lab

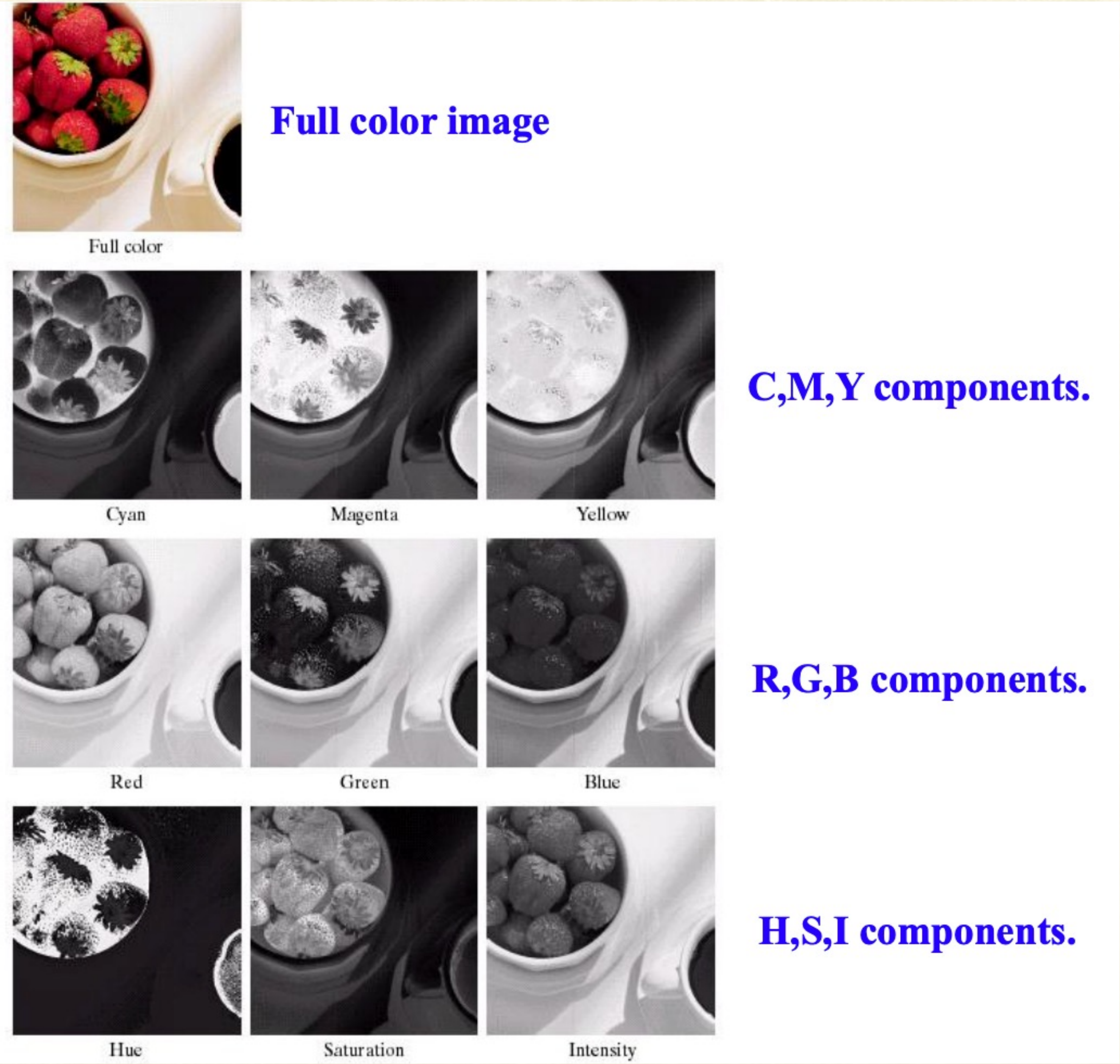


H_{HSV} S_{HSV} V_{HSV}

L^* a^* b^*



• CMY vs RGB vs HSI



Coloring Mughal-e-Azam [1960]

- The colourisation team spent 18 months developing software for colouring the frames, called "Effects Plus”.
- Designed to accept only those colours whose hue would match the shade of grey present in the original film. This ensured that the colours added were as close to the real colour as possible.
- The authenticity of the colouring was later verified when a costume used in the film was retrieved from a warehouse, and its colours were found to closely match those in the film.
- Every shot was finally hand-corrected to perfect the look. The actual colourisation process took a further 10 months to complete in 2004. (cost: Rs. 3.5 cr)

Mayabazar (1957), colorized (2007)

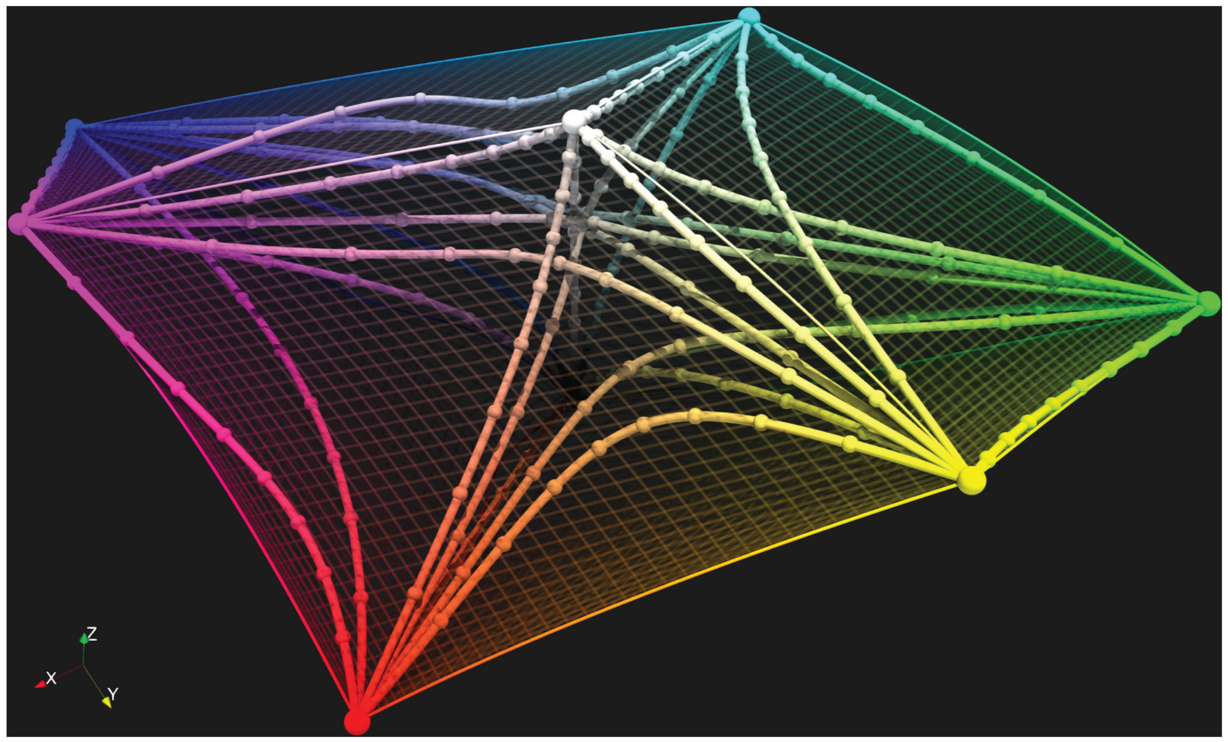
The Film Heritage Foundation announced in March 2015 that they would be restoring *Mayabazar*, along with a few other Indian films from 1931 to 1965, as a part of their restoration projects carried out in India and abroad in accordance with international parameters. **The foundation opposed digital colourisation, stating that they "believe in the original repair as the way the master or the creator had seen it"**

Errors in Perceptual Color Space

- Roxana Bujack, Emily Teti, Jonah Miller, and Terece L. Turton, “The non-Riemannian nature of perceptual color space”, Proceedings of the National Academy of Sciences, (April 2022), DOI: 10.1073/pnas.2119753119.

- Shows that the perceptual color space is probably non-Riemannian:

$$\Delta E(A,B) + \Delta E(B,C) > \Delta E(A,C)$$

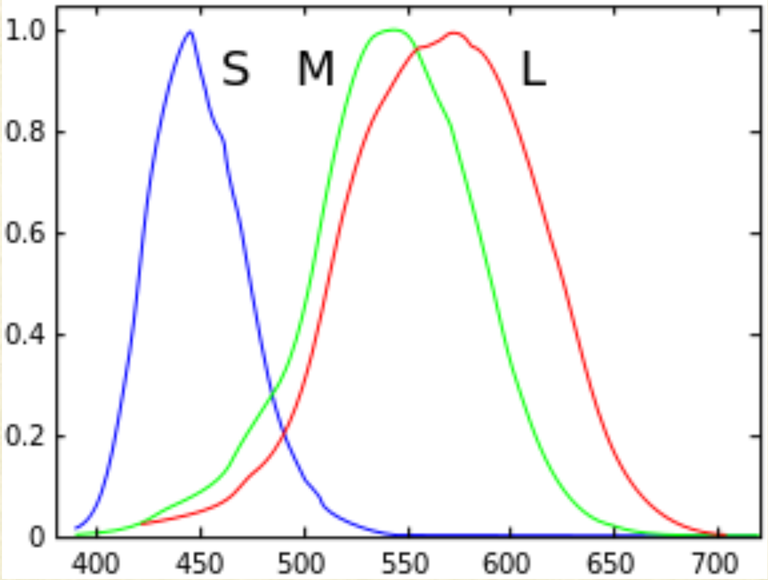


Geodesics in CIE Lab color space: Known since 2000

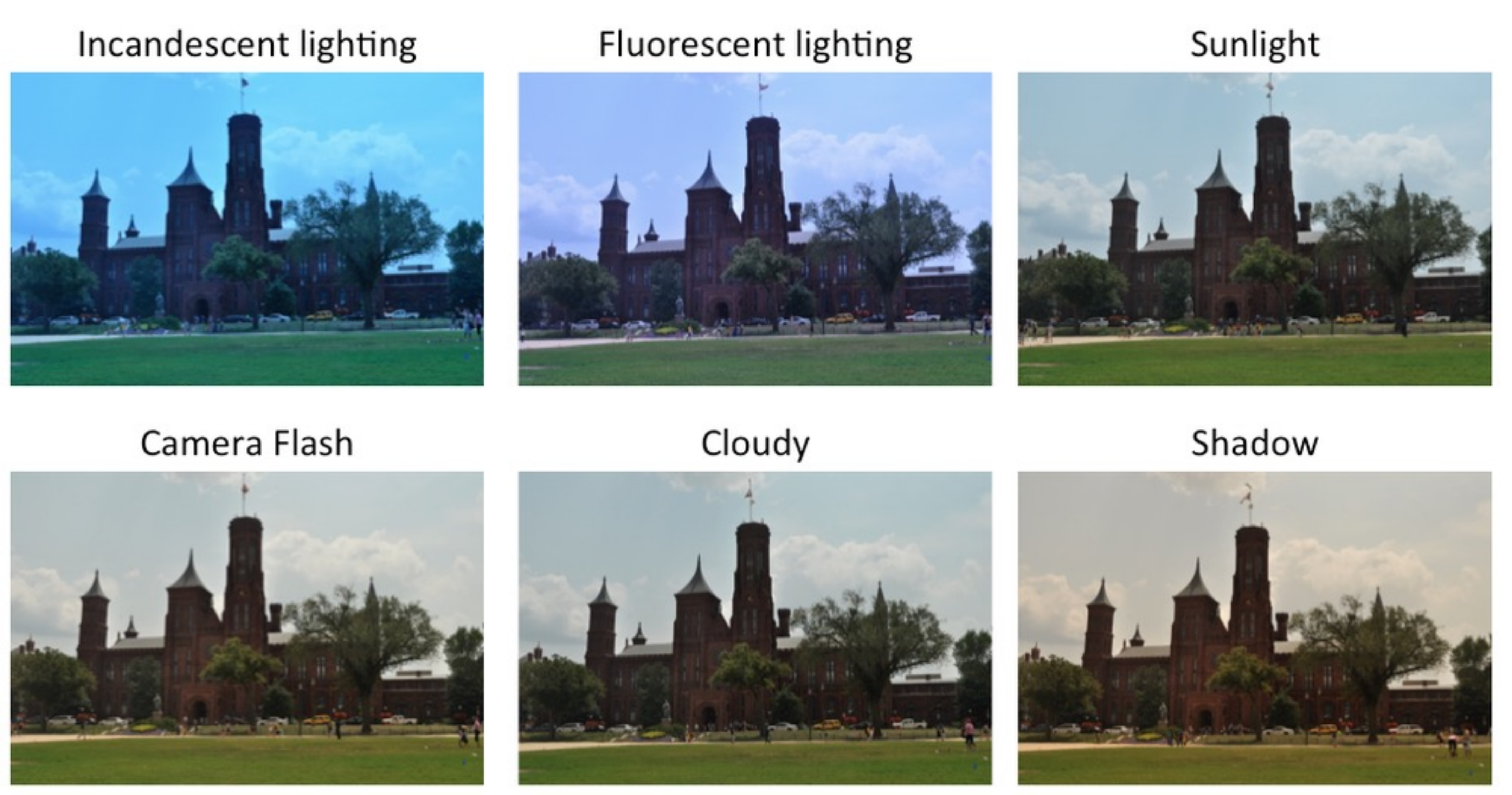
LMS color space

- Used when performing chromatic adaptation (estimating appearance of a sample under a different illuminant)
- Also useful in the study of color blindness, when one or more cone types are defective.

$$\begin{bmatrix} L \\ M \\ S \end{bmatrix} = \begin{bmatrix} 0.7238 & 0.4296 & -0.1624 \\ -0.7036 & 1.6975 & 0.0061 \\ 0.0030 & 0.0136 & 0.9834 \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \end{bmatrix}$$



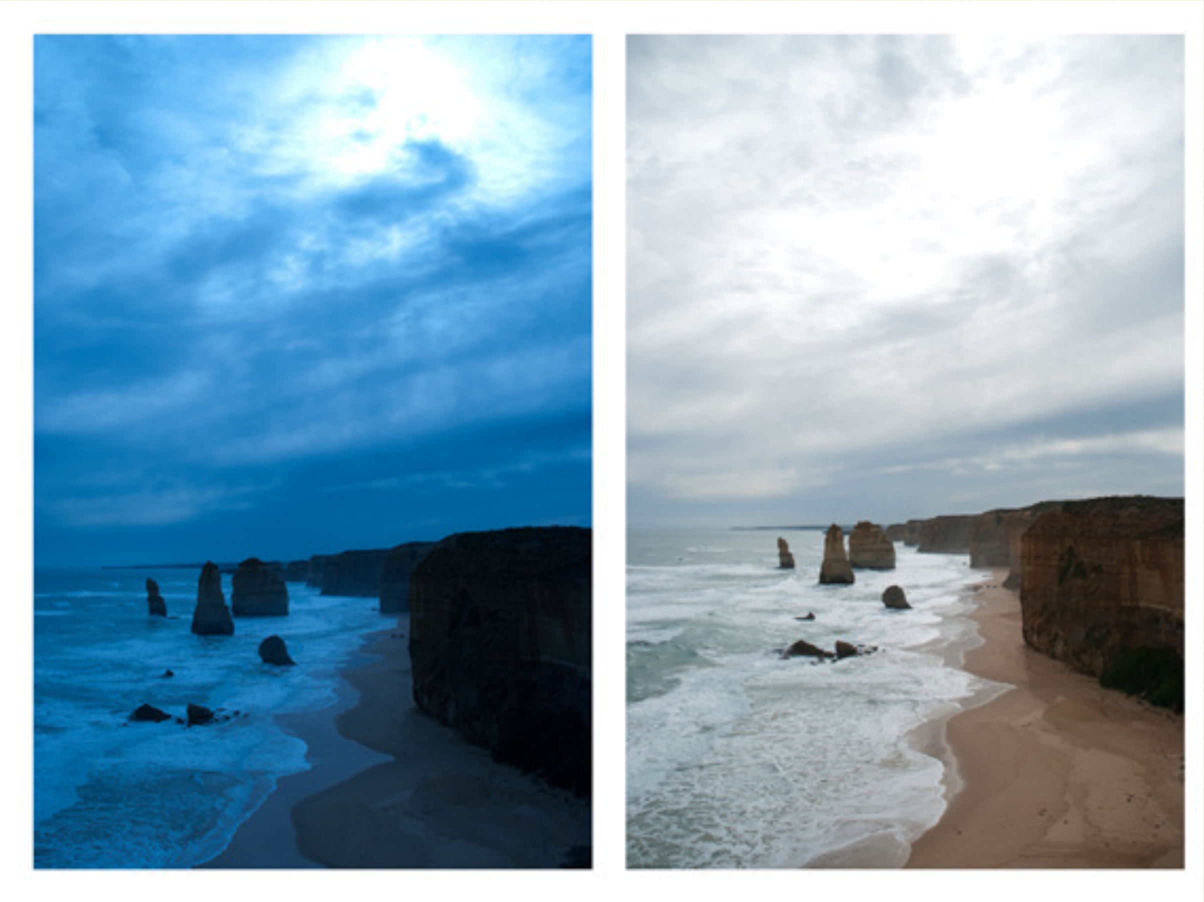
White Balancing



courtesy: seventhgrove.com

White Balancing

Courtesy: wahlmanphotography.com



Courtesy: wallcreations.com.au

White Balancing

White Balancing

- Use reference cards
  - White, 50% gray, etc.

White Balancing

```
im = double(imread('lighthouse.jpg'));

RGBw = [246 169 87];

im1(:,:,1) = im(:,:,1)*255/RGBw(1);
im1(:,:,2) = im(:,:,2)*255/RGBw(2);
im1(:,:,3) = im(:,:,3)*255/RGBw(3);
```

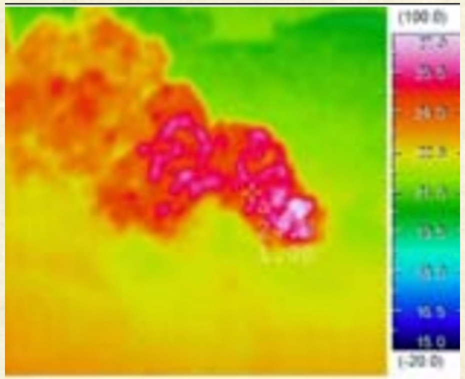
- Von Kries Method
  - White Balancing in LMS color space

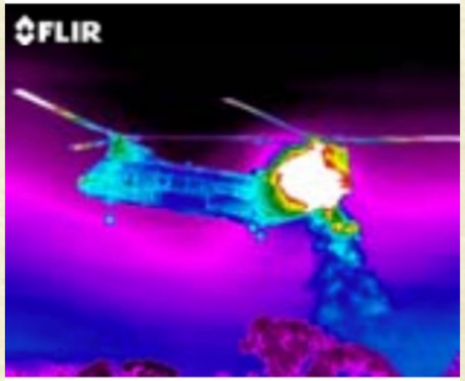
$$\begin{bmatrix} L \\ M \\ S \end{bmatrix} = \begin{bmatrix} 1/L'_w & 0 & 0 \\ 0 & 1/M'_w & 0 \\ 0 & 0 & 1/S'_w \end{bmatrix} \begin{bmatrix} L' \\ M' \\ S' \end{bmatrix}$$

The von Kries coefficient rule is based on the assumption that color constancy is achieved by individually adapting the gains of the three cone responses

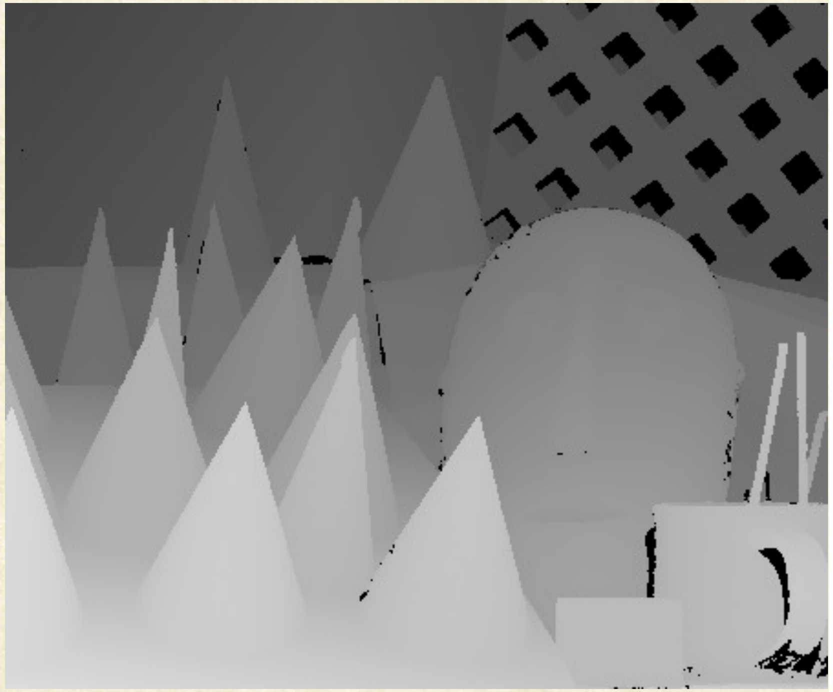
Pseudocoloring of Images

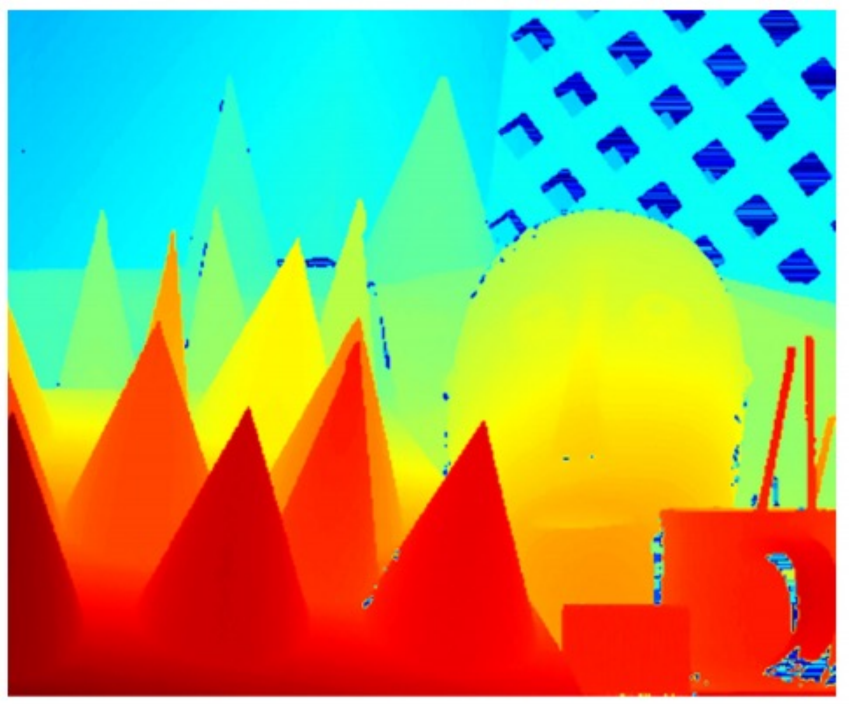
- Pseudocoloring or false coloring consists of assigning colors to grey values based on any defined criterion.
- Primary goal is improving human visualization
  - Humans can discern between thousands of colors+intensities as opposed to around 2 dozed shades of grey.





Pseudo color Image Processing



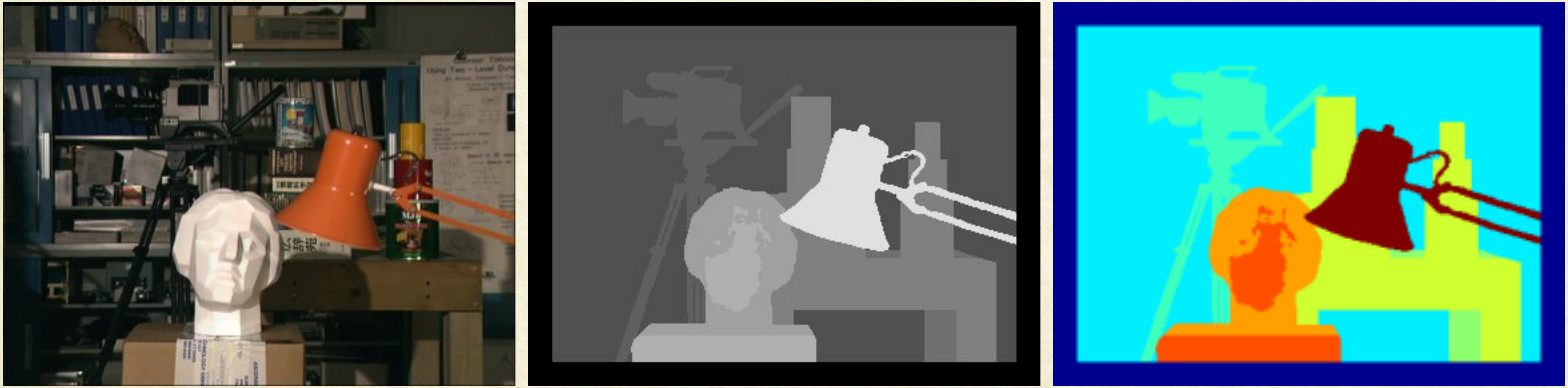


Colors to indicate depth

Courtesy: Middlebury dataset

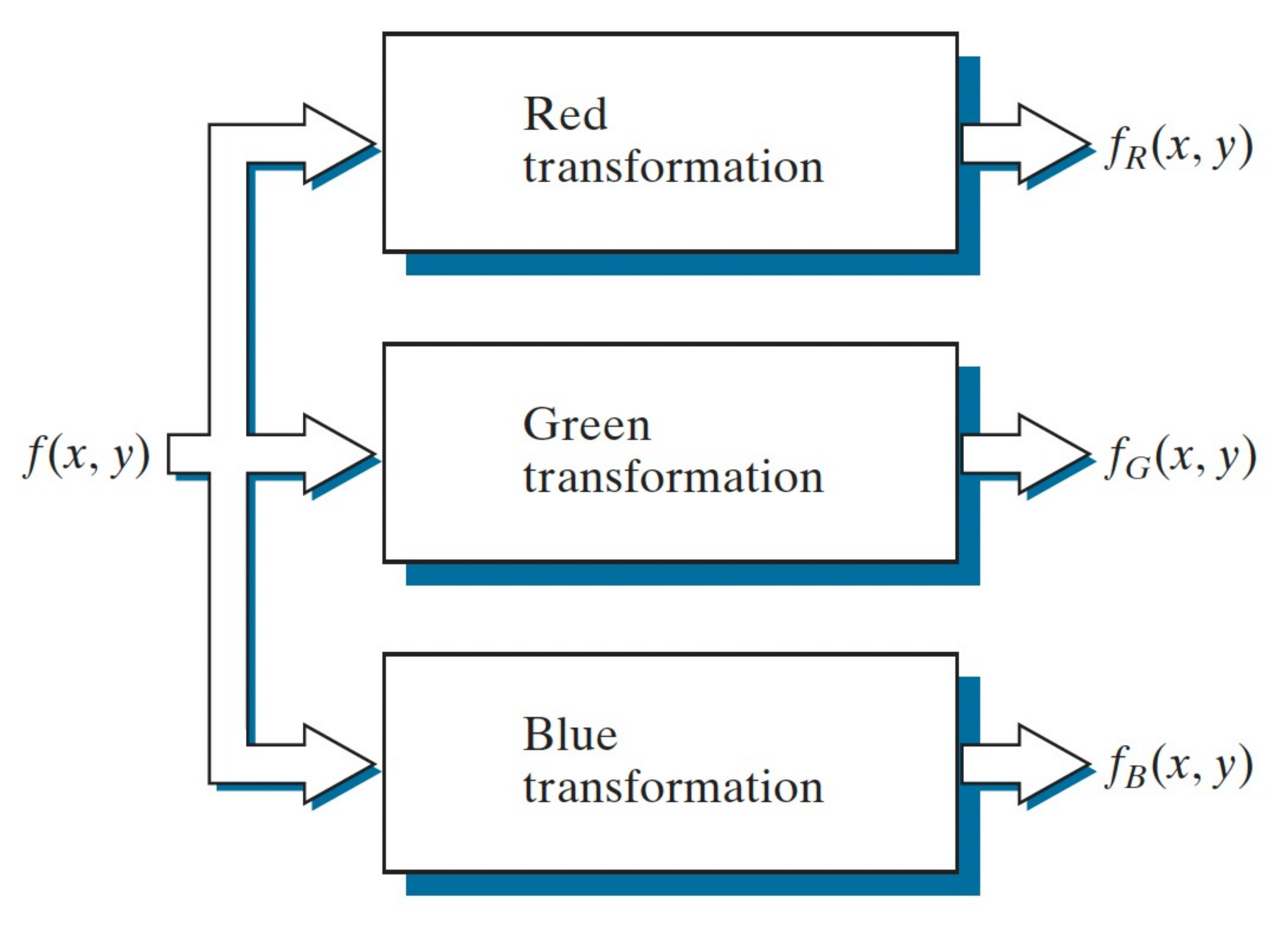
Pseudo color Image Processing

- Colors to indicate Labels



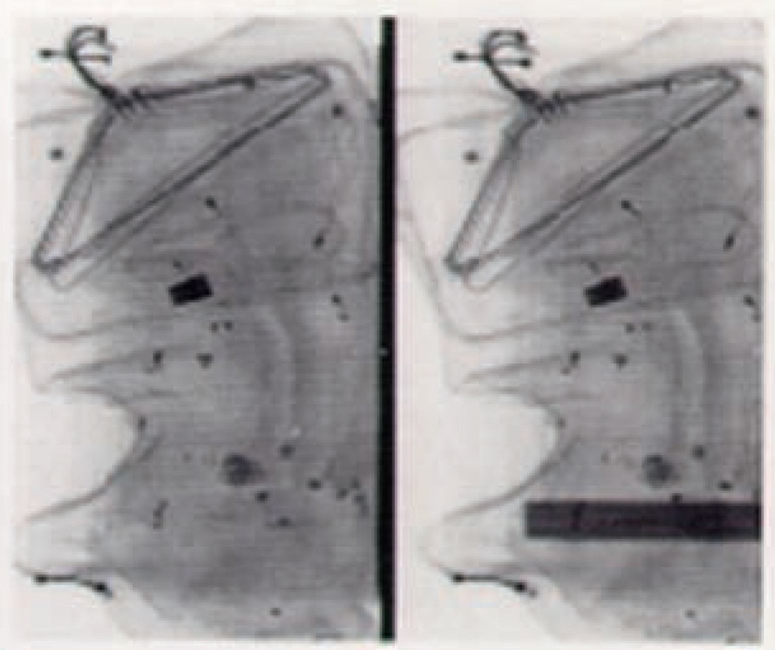
courtesy: middlebury dataset

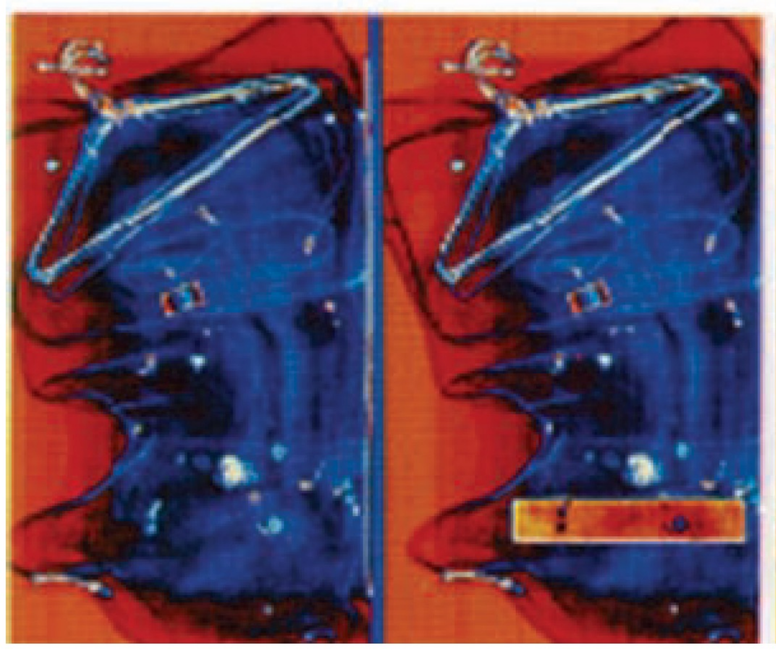
Pseudo color Image Processing (Transformations)

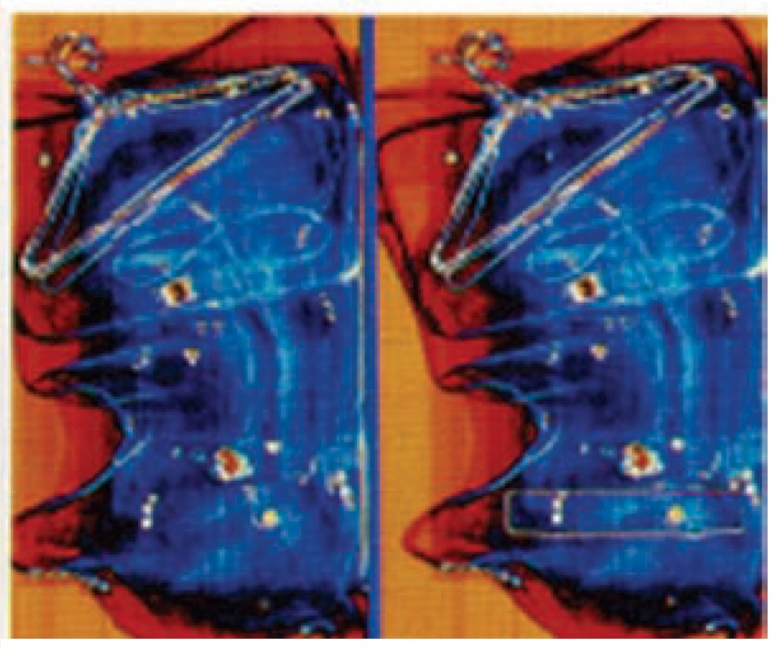


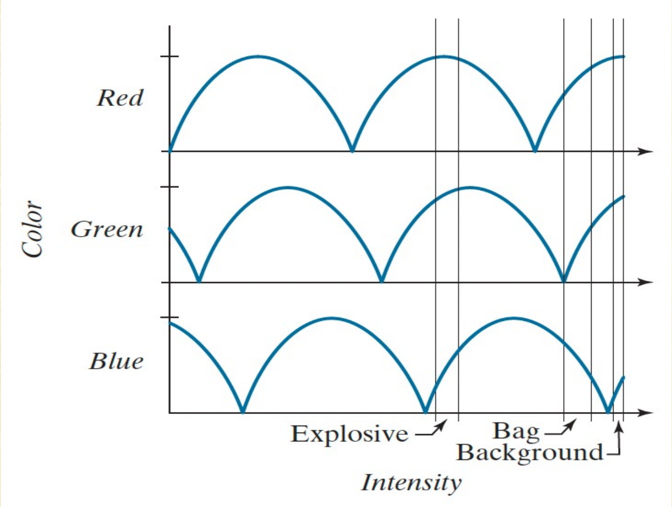
Courtesy: Gonzalez and Woods

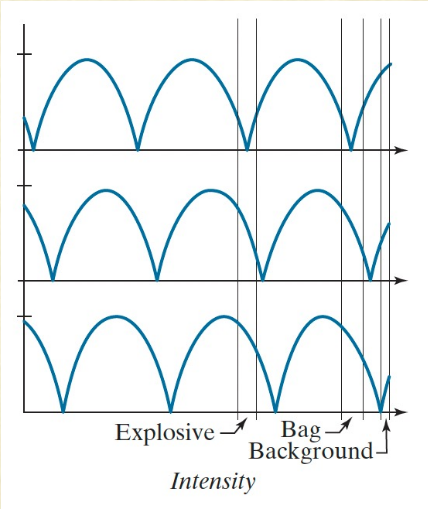
Pseudo color Image Processing (Transformations)





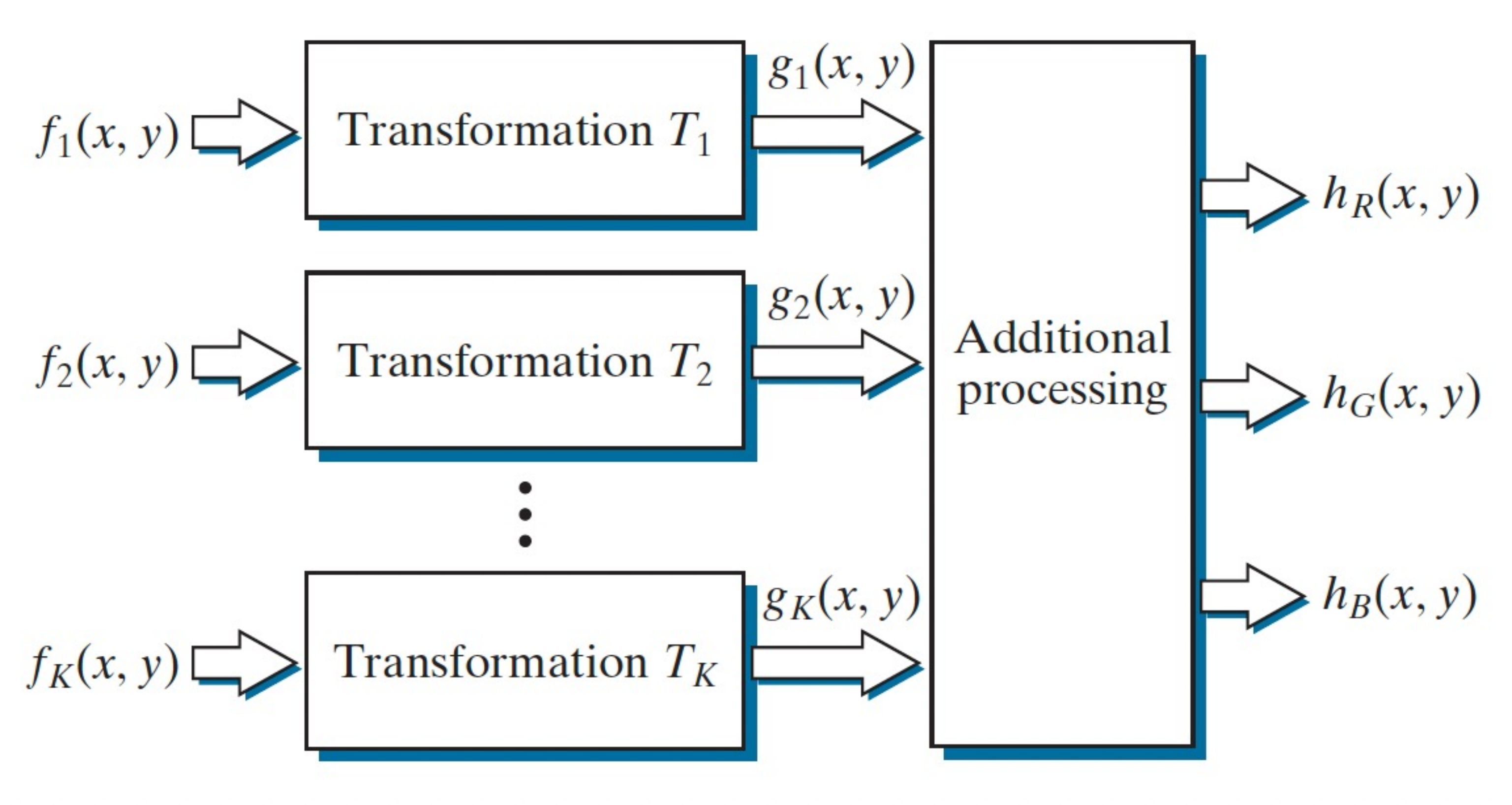


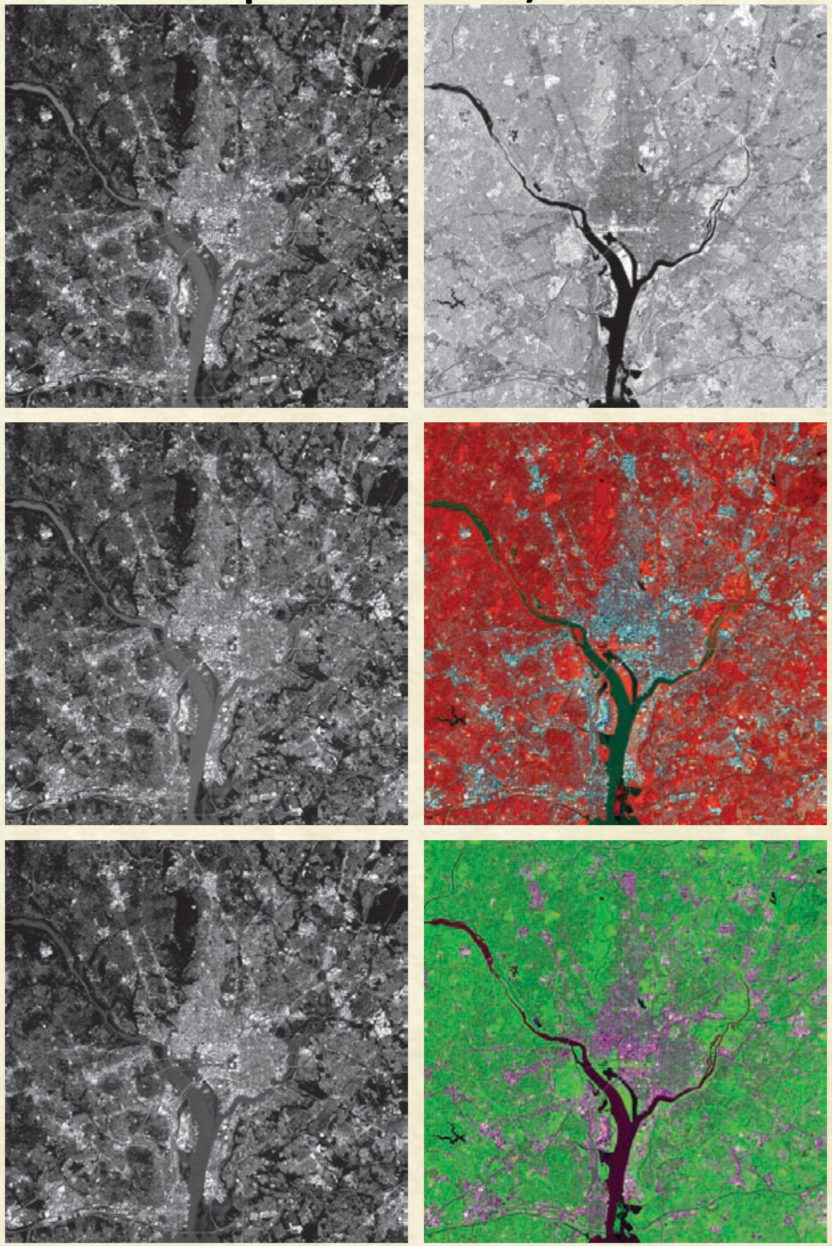




Courtesy: Gonzalez and Woods

Pseudo color Image Processing (Multi Spectral)

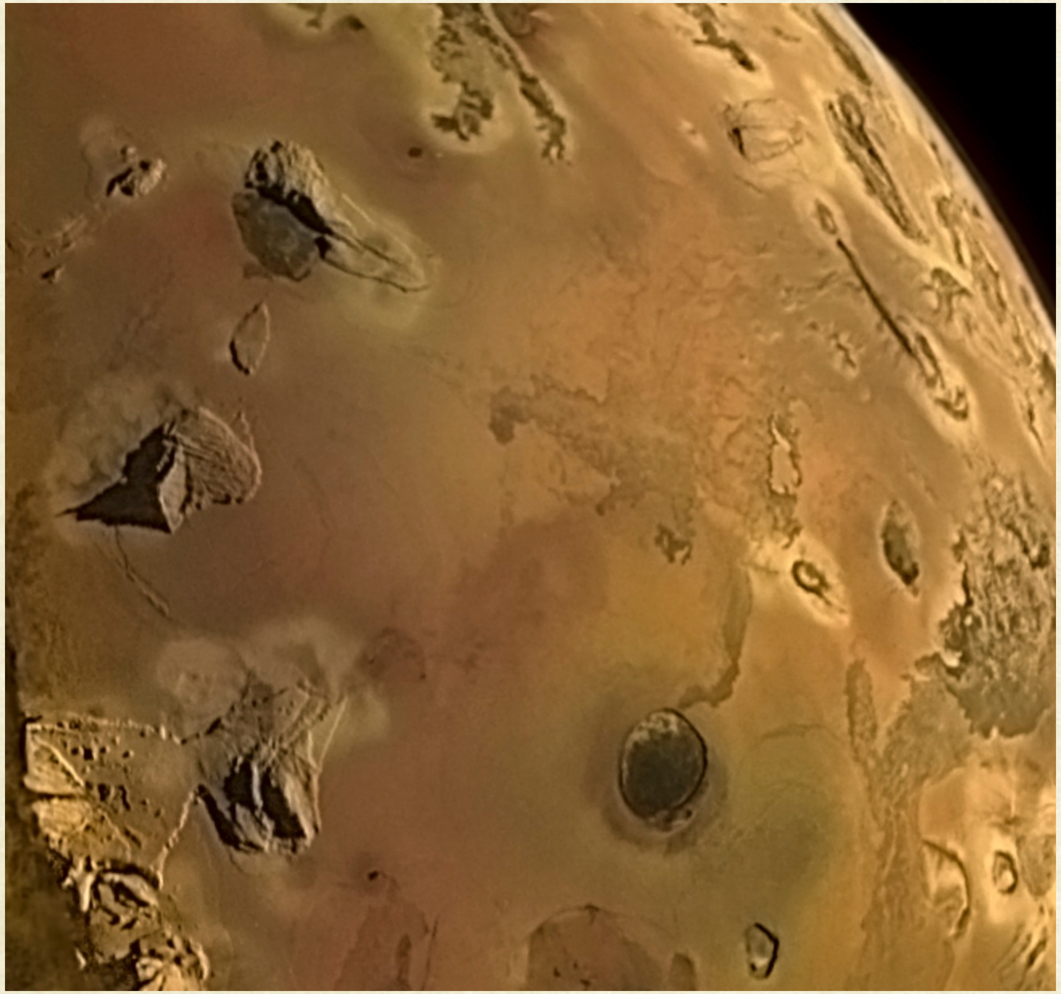




R	IR
G	IR, G, B
B	R, IR, B

Courtesy: Gonzalez and Woods

Pseudo color Image Processing (Multi Spectral)

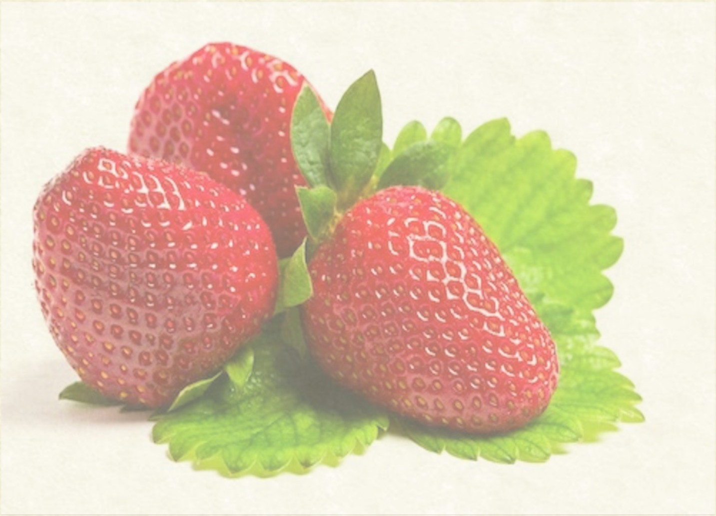


(a) High-resolution image of Jupiter’s moon, Io, captured by the Juno satellite from NASA (and rendered in color) and (b) another pseudo-color image created by a different processing of the captured raw data.

RGBA space

- A (alpha) for transparency (important in image editing)

$$I_{out} = \alpha \, I_{foreground} + (1 - \alpha) \, I_{background}$$



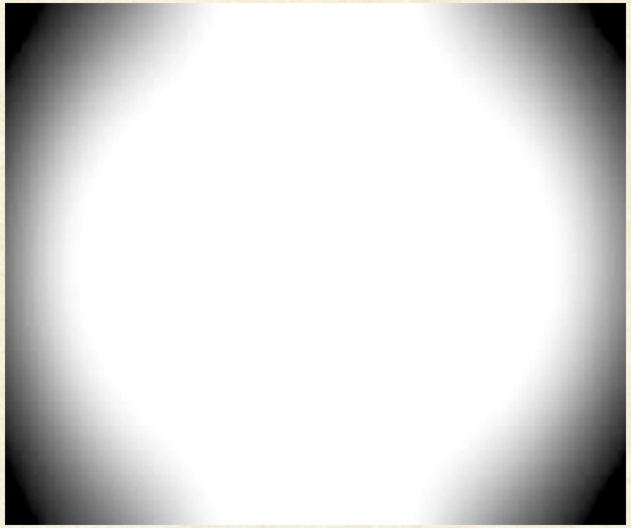
Applications: Image enhancement in RGB

Example: Vintage effect or Sepia Toning

$$\begin{bmatrix} R_{sep} \\ G_{sep} \\ B_{sep} \end{bmatrix} = \begin{bmatrix} 0.293 & 0.769 & 0.210 \\ 0.249 & 0.686 & 0.188 \\ 0.172 & 0.534 & 0.151 \end{bmatrix} \begin{bmatrix} R \\ G \\ B \end{bmatrix}$$

Example: Vignetting effect

$\times$



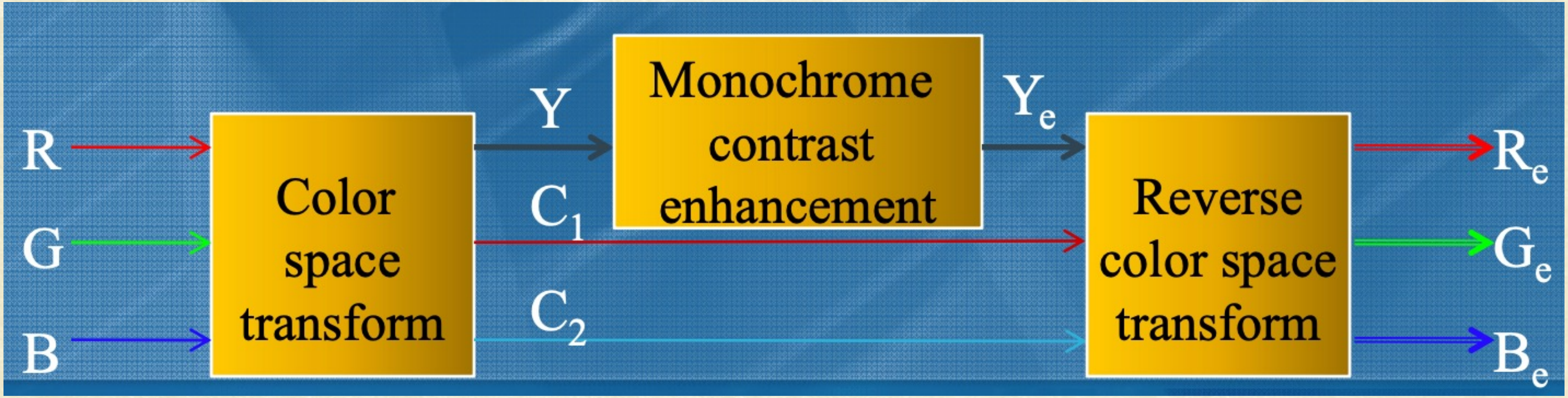
$=$

```
txt = imread(texture_path);
txt = imresize(txt,[size(out,1) size(out,2)]);
txt = double(rgb2gray(txt))/255;
```

```
out1(:,:,1) = double(out(:,:,1)) .* double(txt);
out1(:,:,2) = double(out(:,:,2)) .* double(txt);
out1(:,:,3) = double(out(:,:,3)) .* double(txt);
```

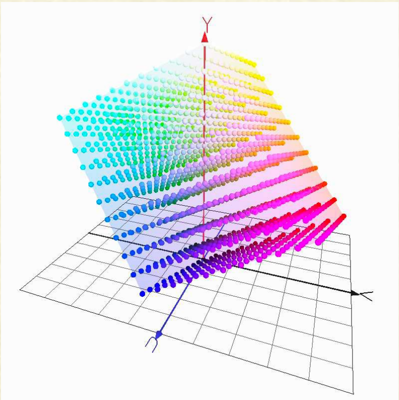
Contrast Enhancement in Color Image

- Basic (popular) Approach:
  - Human eye is more sensitive to brightness contrast than color contrast
  - Can achieve good contrast enhancement on brightness component alone

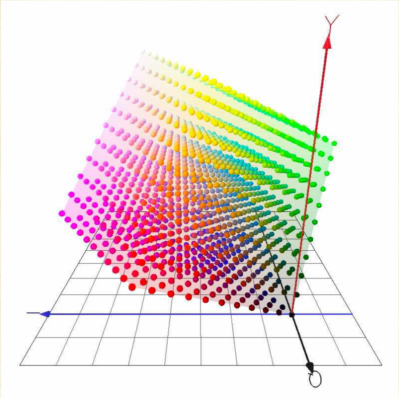


YC_1C_2 color spaces

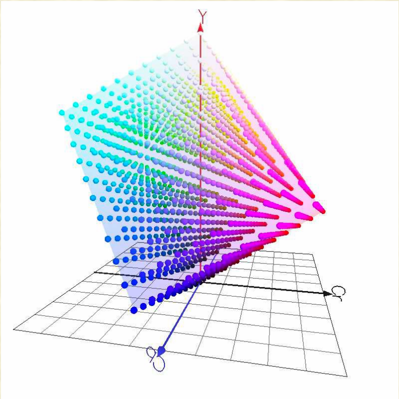
- $$\begin{bmatrix} Y \\ U \\ V \end{bmatrix} = \begin{bmatrix} 0.299 & 0.587 & 0.114 \\ -0.147 & -0.289 & 0.436 \\ 0.615 & -0.515 & -0.100 \end{bmatrix} \begin{bmatrix} R \\ G \\ B \end{bmatrix}$$



- $$\begin{bmatrix} Y \\ I \\ Q \end{bmatrix} = \begin{bmatrix} 0.299 & 0.587 & 0.114 \\ 0.596 & -0.274 & -0.322 \\ 0.212 & -0.523 & 0.311 \end{bmatrix} \begin{bmatrix} R \\ G \\ B \end{bmatrix}$$



- $$\begin{bmatrix} Y \\ C_b \\ C_r \end{bmatrix} = \begin{bmatrix} 0.2989 & 0.5866 & 0.1145 \\ -0.1688 & -0.3312 & 0.5000 \\ 0.5000 & -0.4184 & -0.0816 \end{bmatrix} \begin{bmatrix} R \\ G \\ B \end{bmatrix}$$



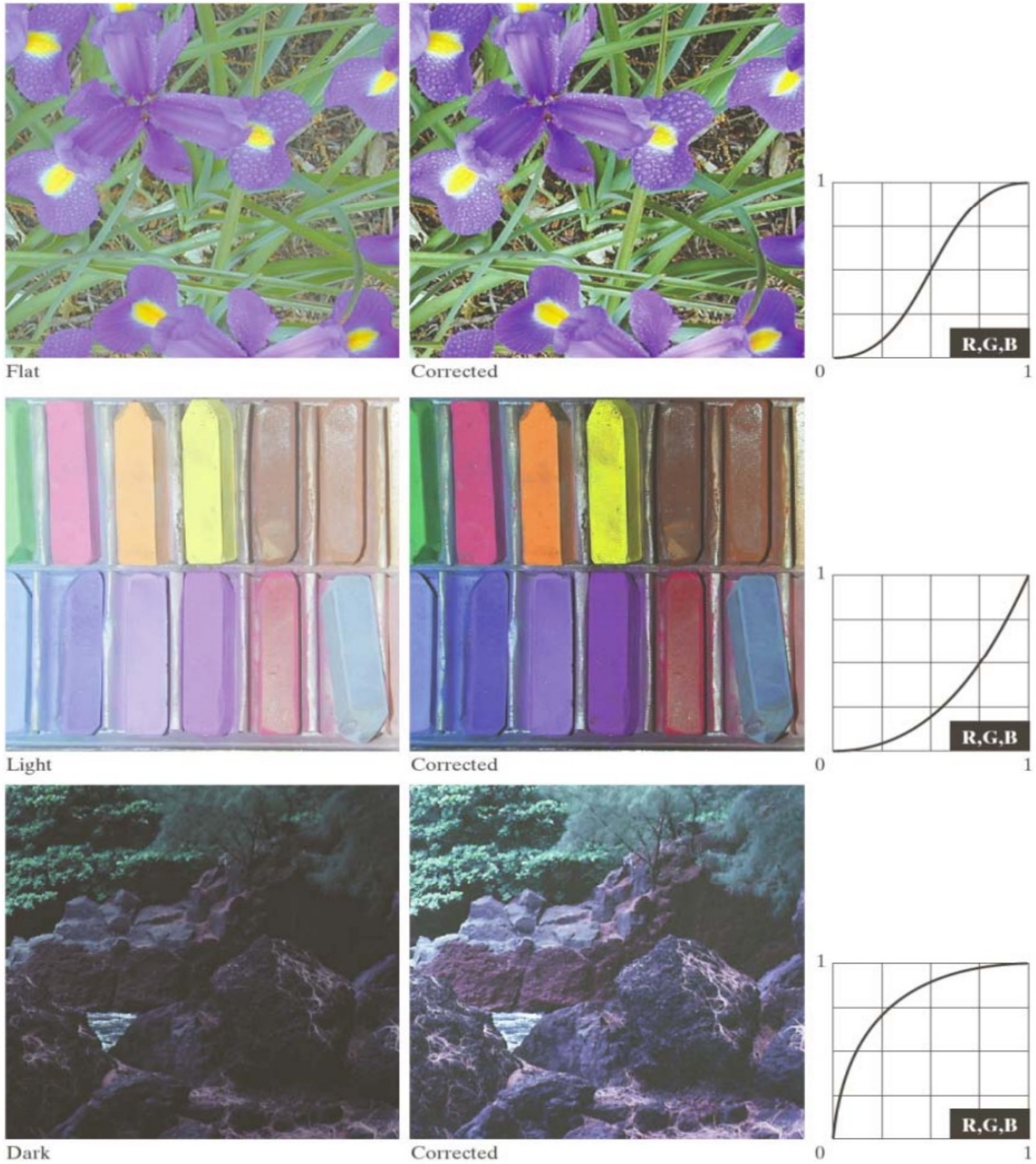
Tone and Color Corrections

$$L^{*} = 116h(\frac{Y}{Y_W}) - 16$$

$$a^{*} = 500[h(\frac{X}{X_W}) - h(\frac{Y}{Y_W})]$$

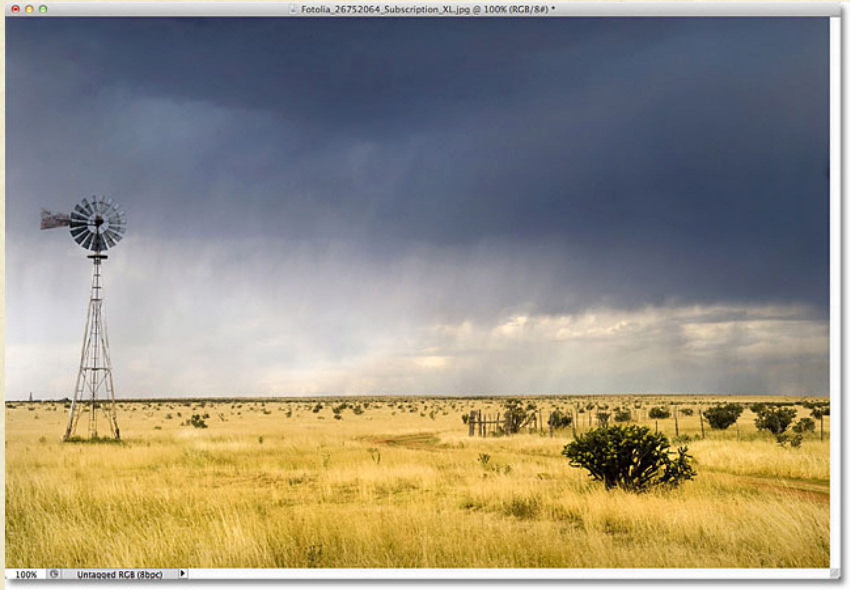
$$b^{*} = 200[h(\frac{Y}{Y_W}) - h(\frac{Z}{X_W})]$$

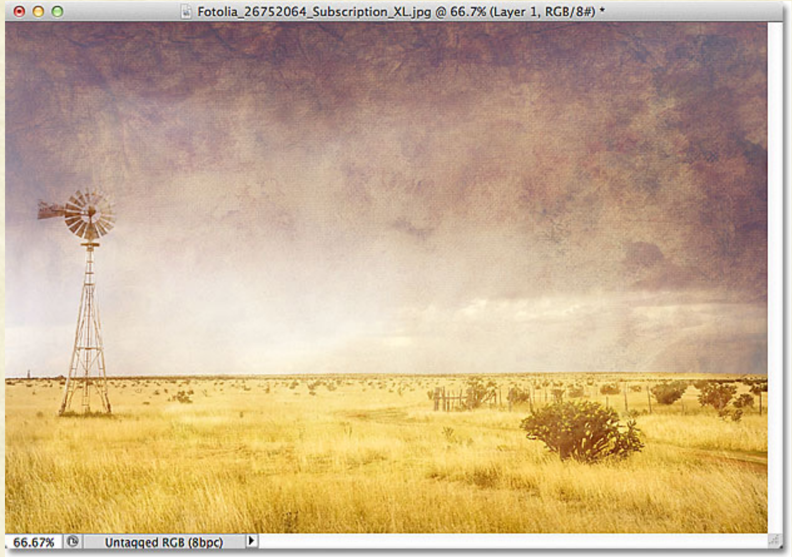
$$h(q) = \begin{cases} \sqrt[3]{q}0.5 & q > 0.008856 \\ 7.787q + 16/116 & q \leq 0.008856 \end{cases}$$



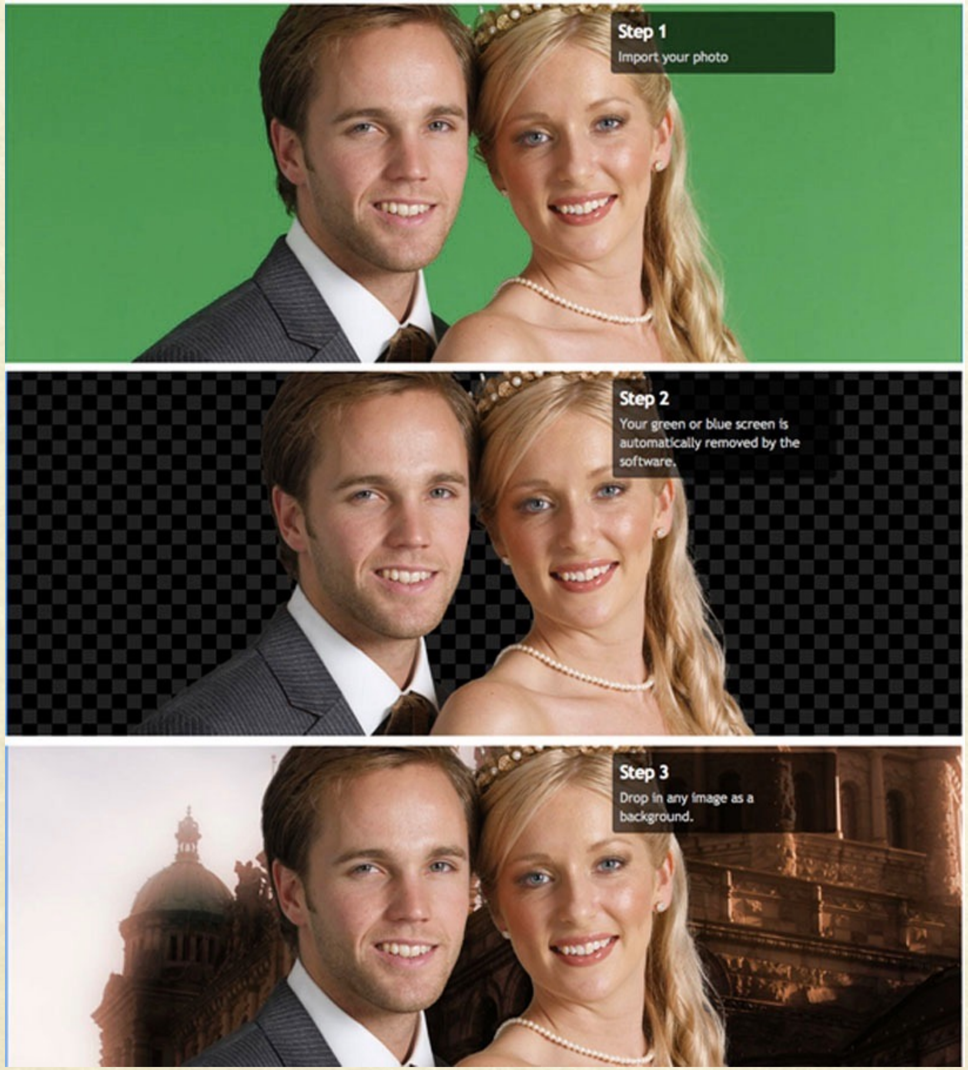
Other such Image Effects

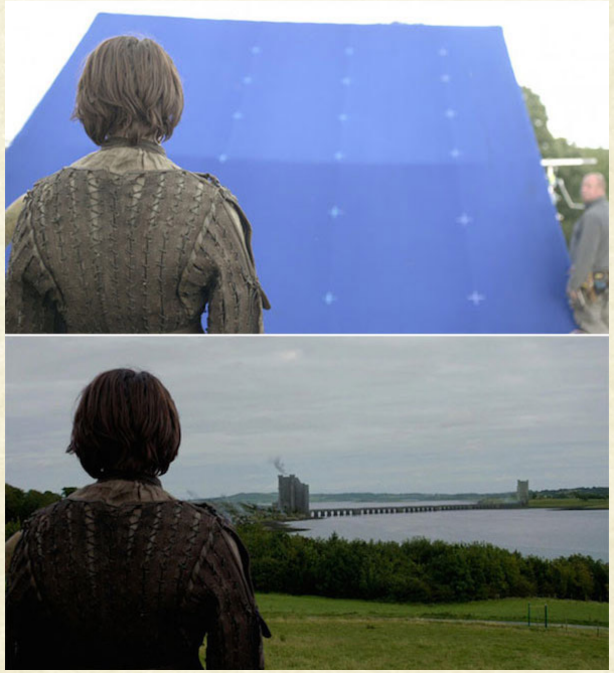
1. Change the transformation matrix, to suit the desired color tones
2. Choose or design different textures and blend them with original image
3. Repeat 1 and 2 in innovative ways



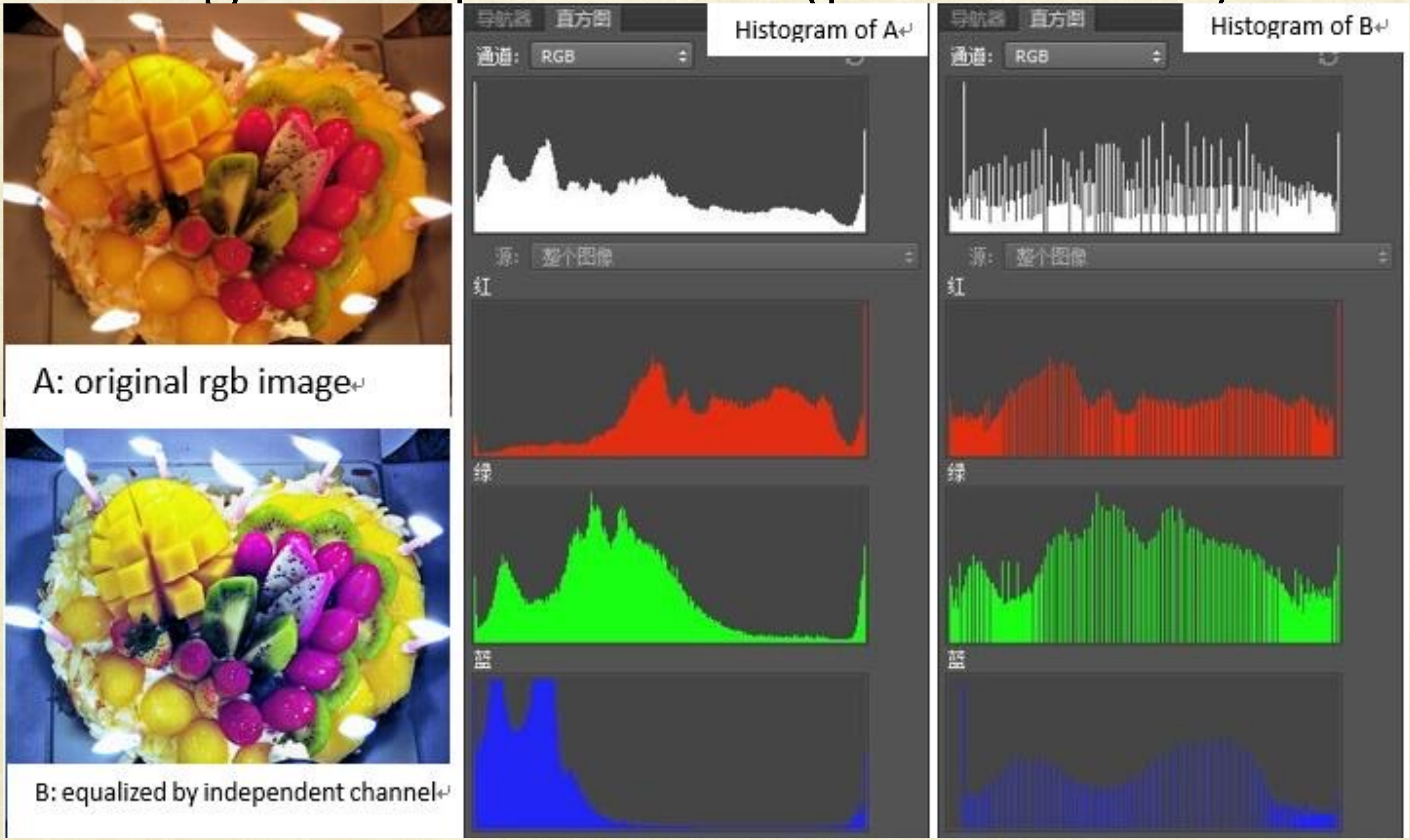


Chroma Keying

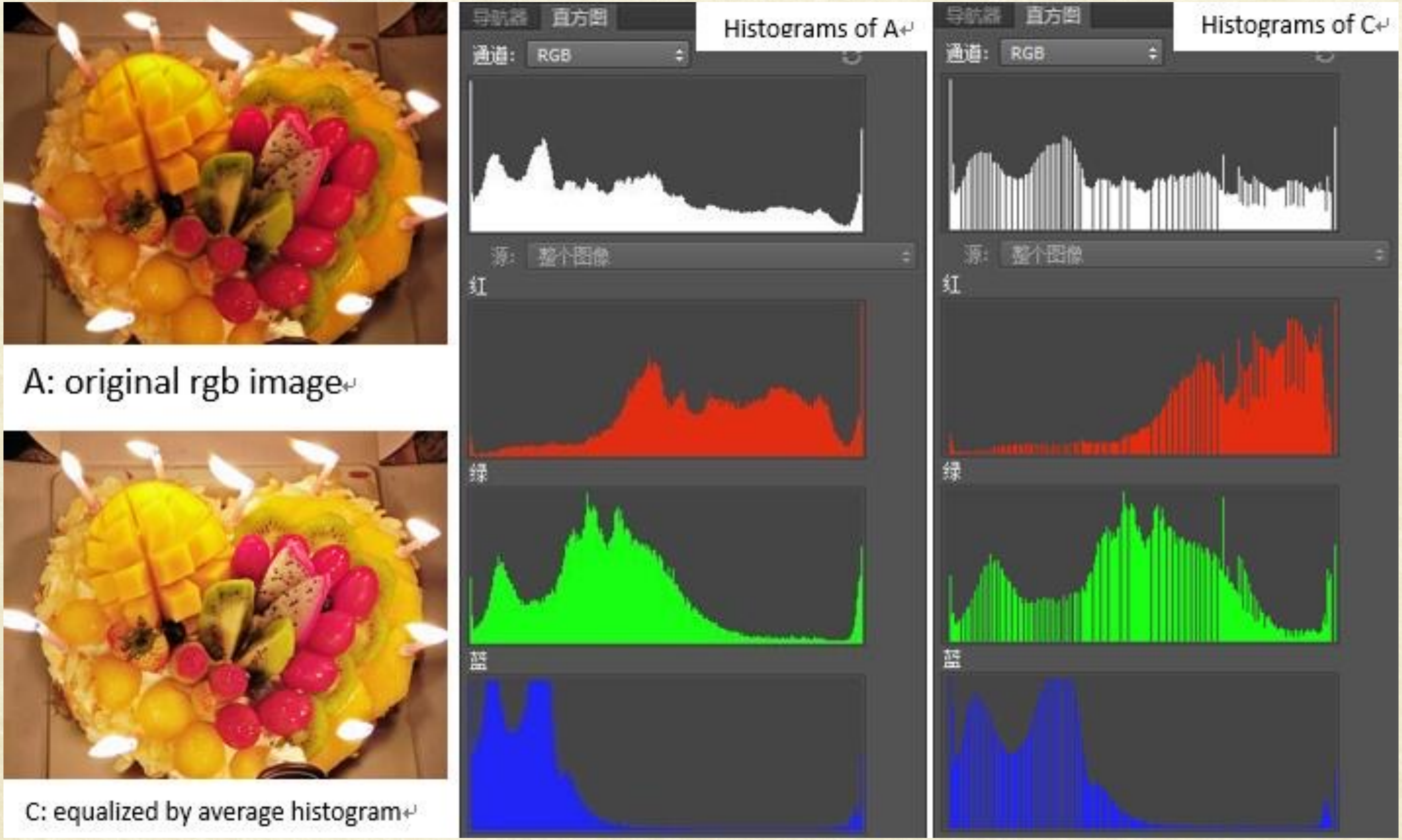




Histogram equalization (per-channel)



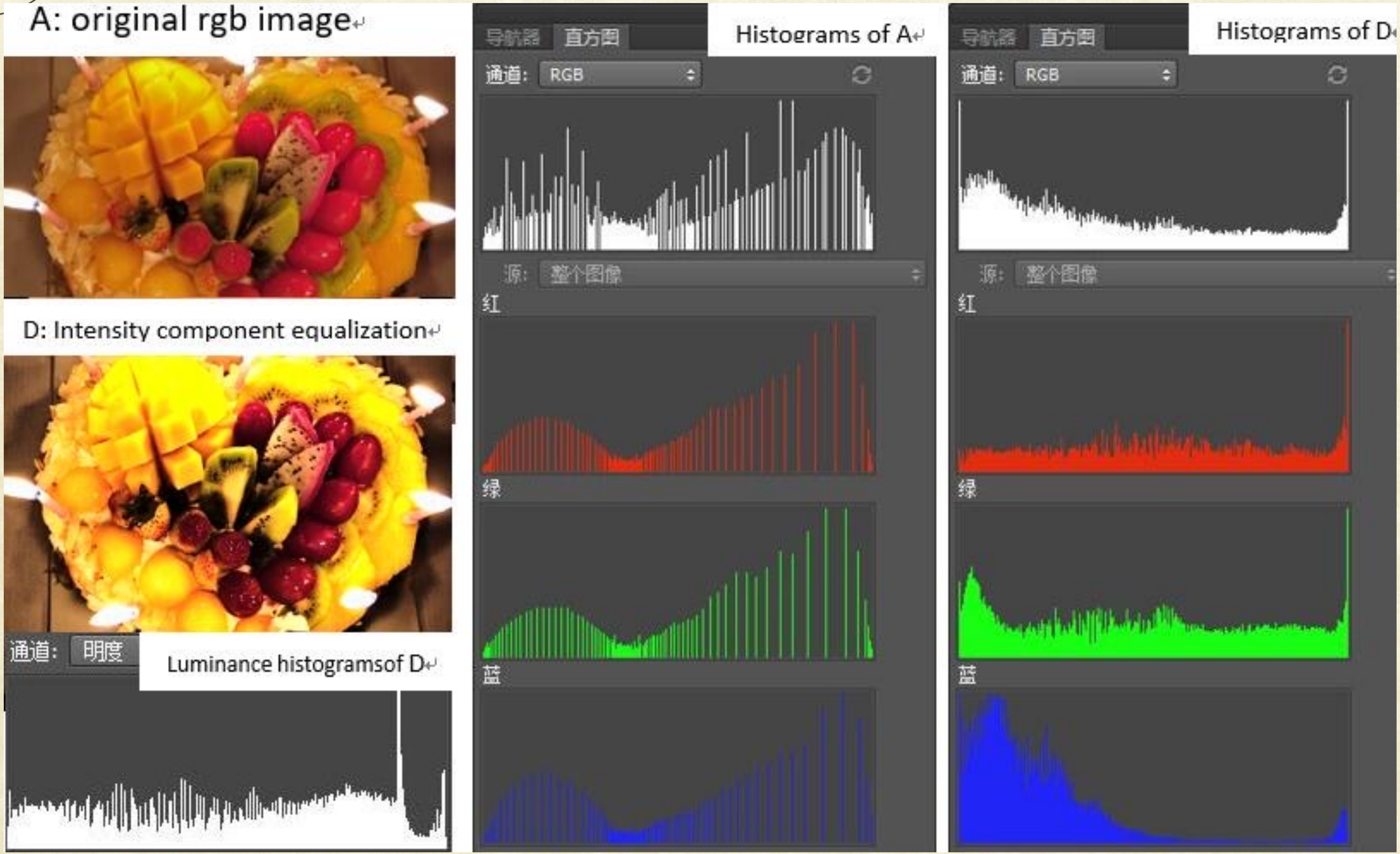
Histogram equalization (ref histogram = average of R,G,B)

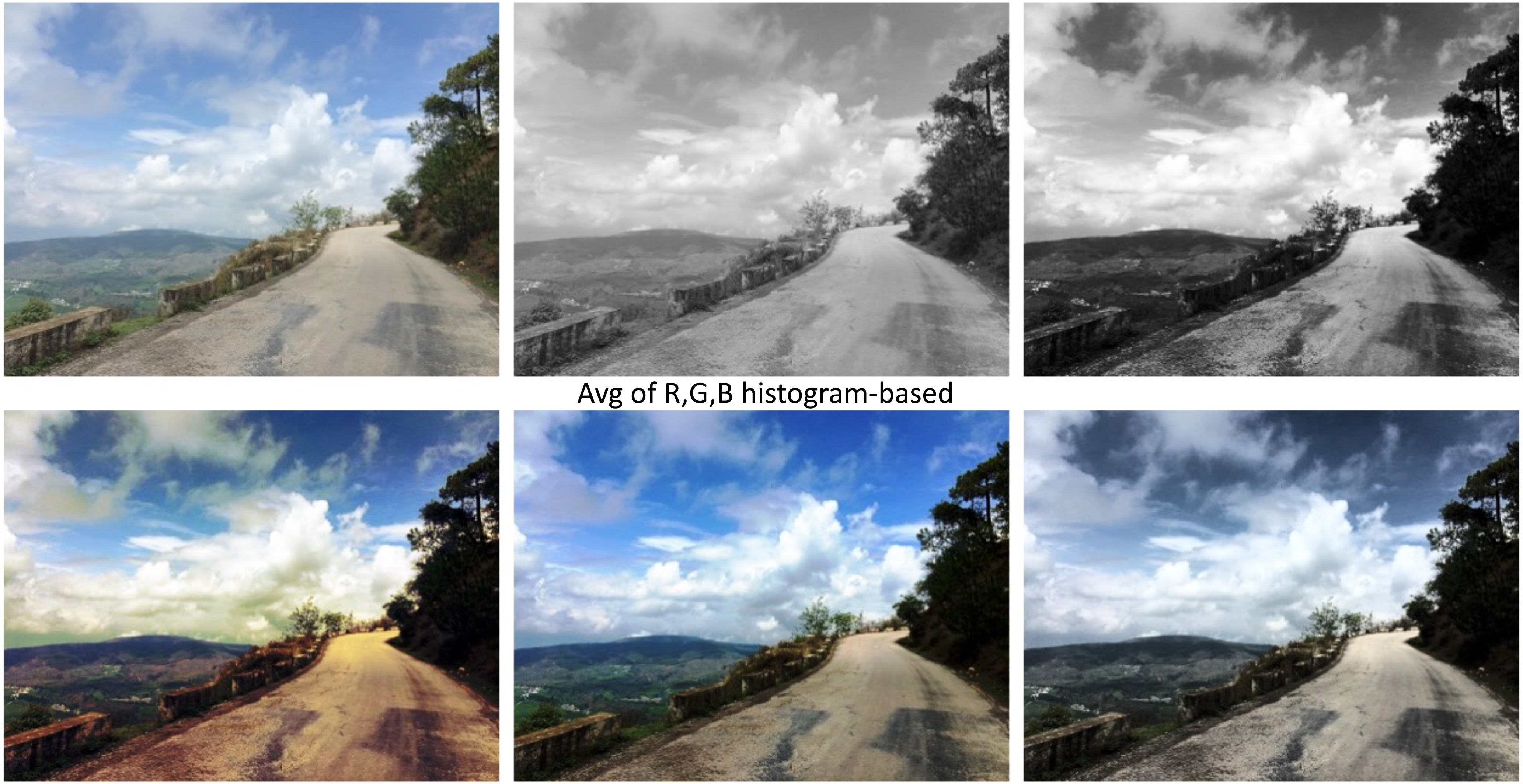


A: original rgb image

C: equalized by average histogram

Histogram equalization (ref histogram = I of HSI)





Avg of R,G,B histogram-based

Independent

Background and foreground that are both bright.

I of HSI

Independent

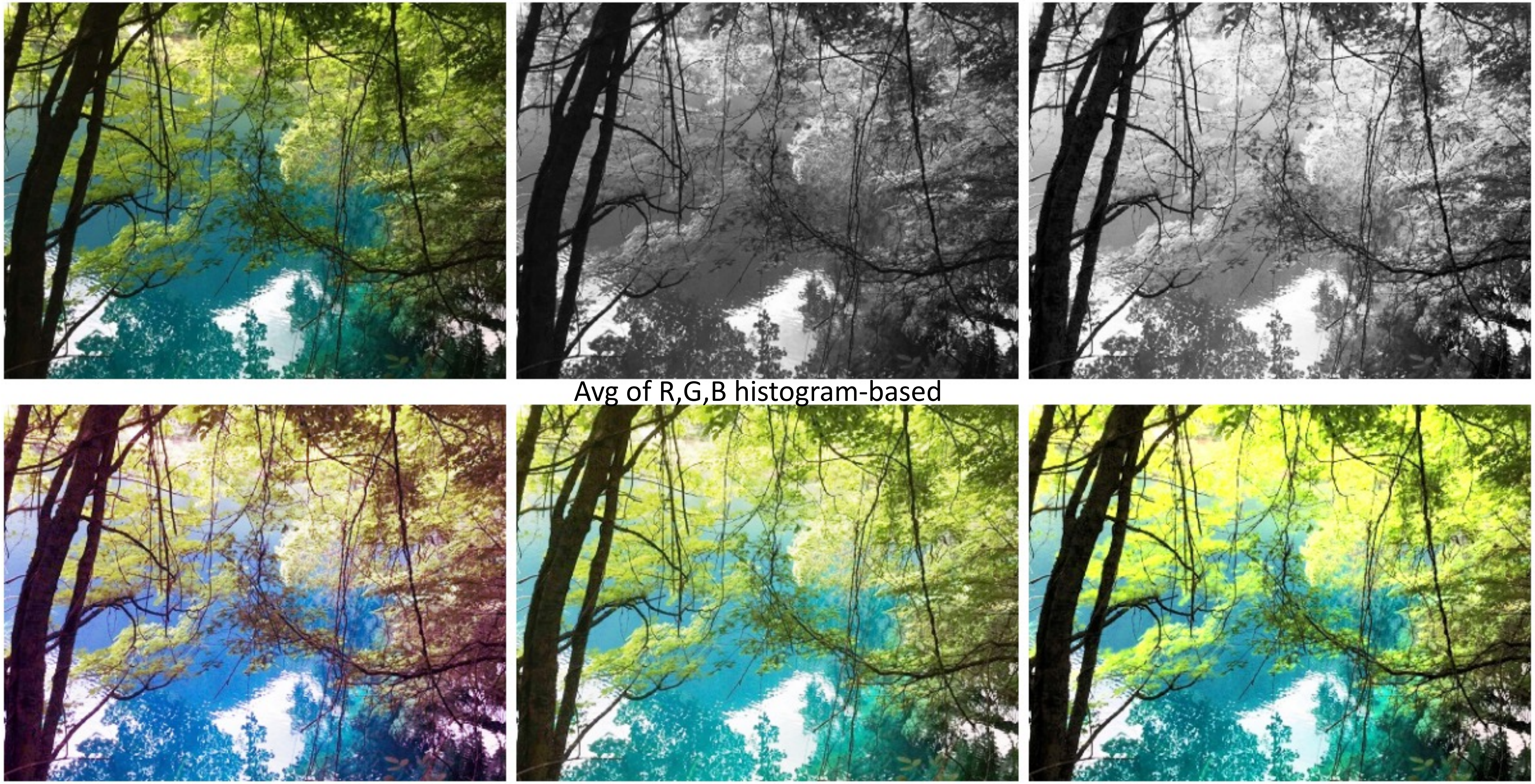
Foreground brighter.

I of HSI

Independent

Lanterns are brighter.

I of HSI



Avg of R,G,B histogram-based

Independent

Local areas are brighter.

I of HSI

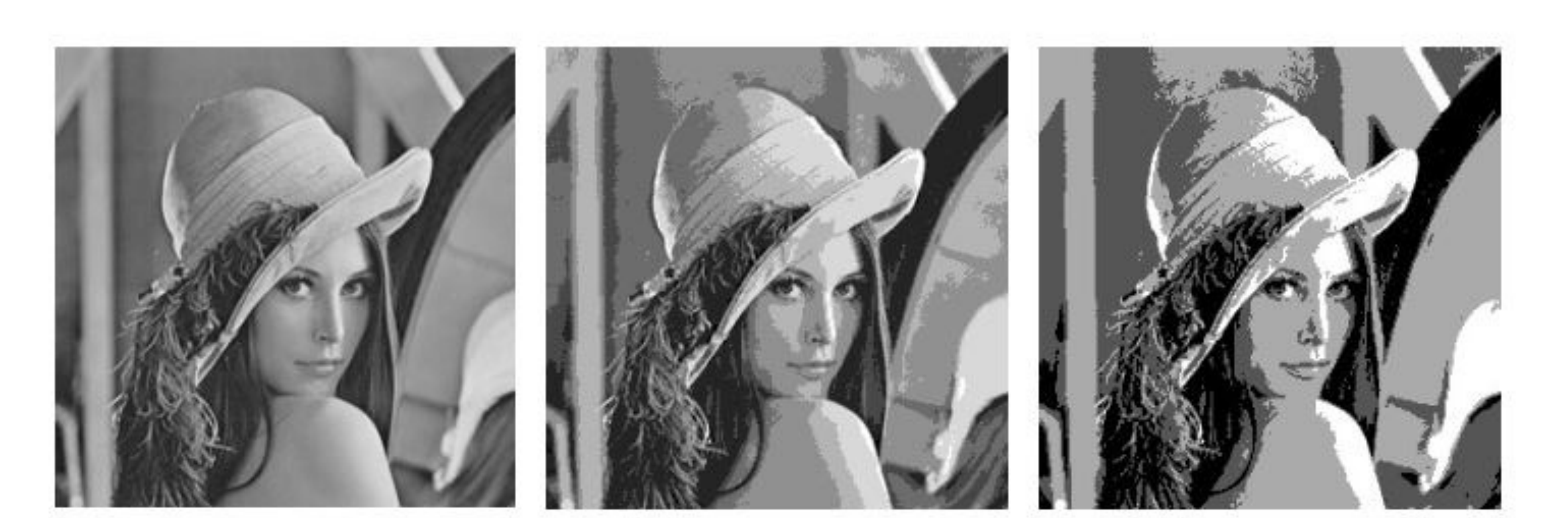
Avg of R,G,B histogram-based

Independent

The grasses are unrealistic and the figure is outlined against the background with white line.

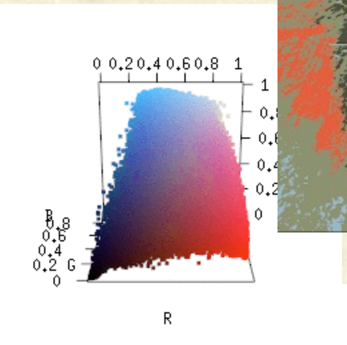
I of HSI

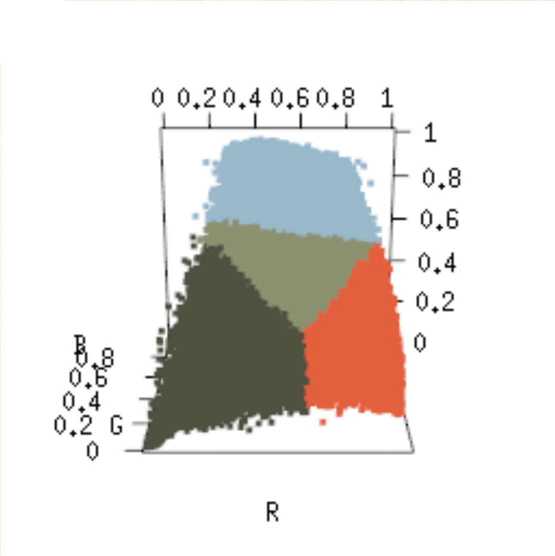
Grayscale Quantization



Original 8 levels 4 levels

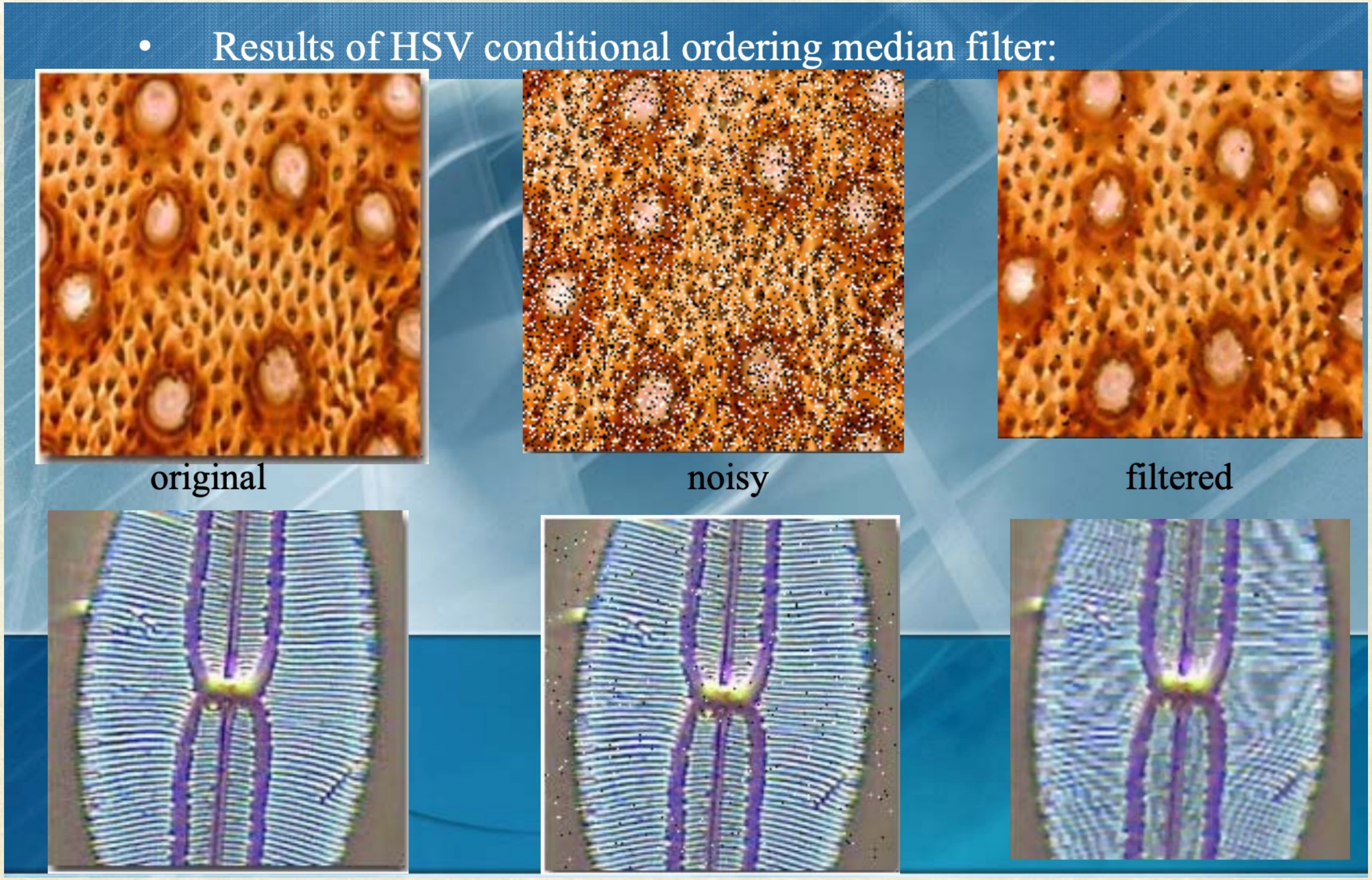
Color Quantization (k-means on RGB)



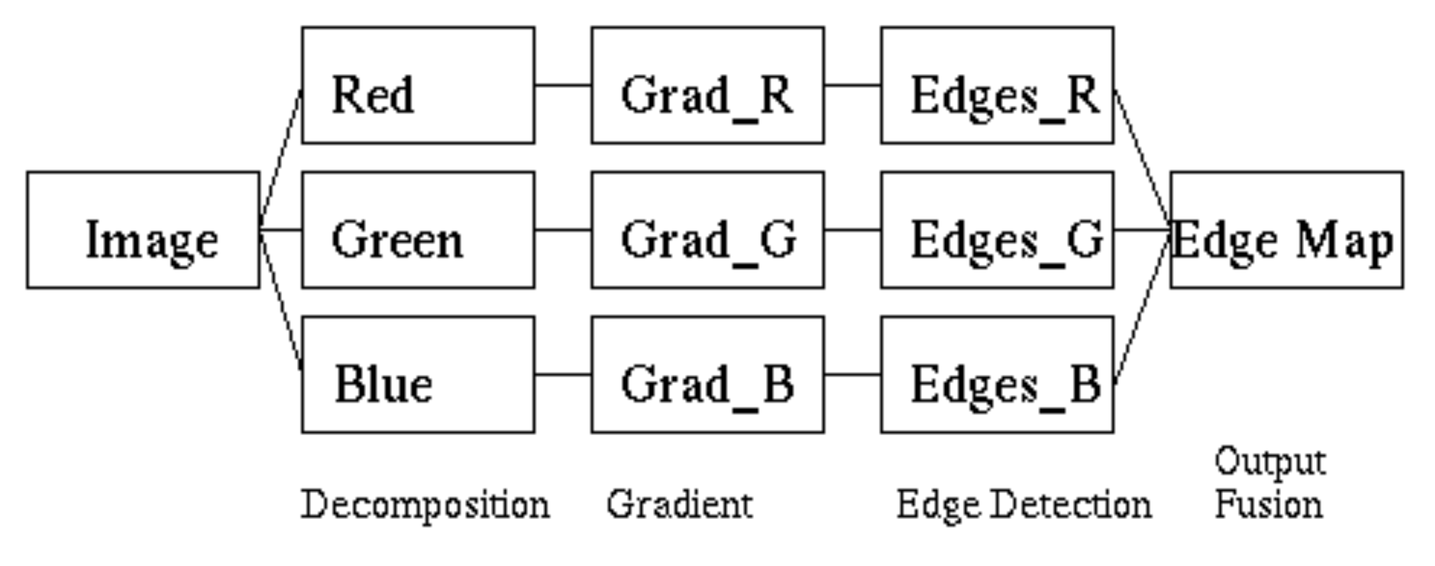


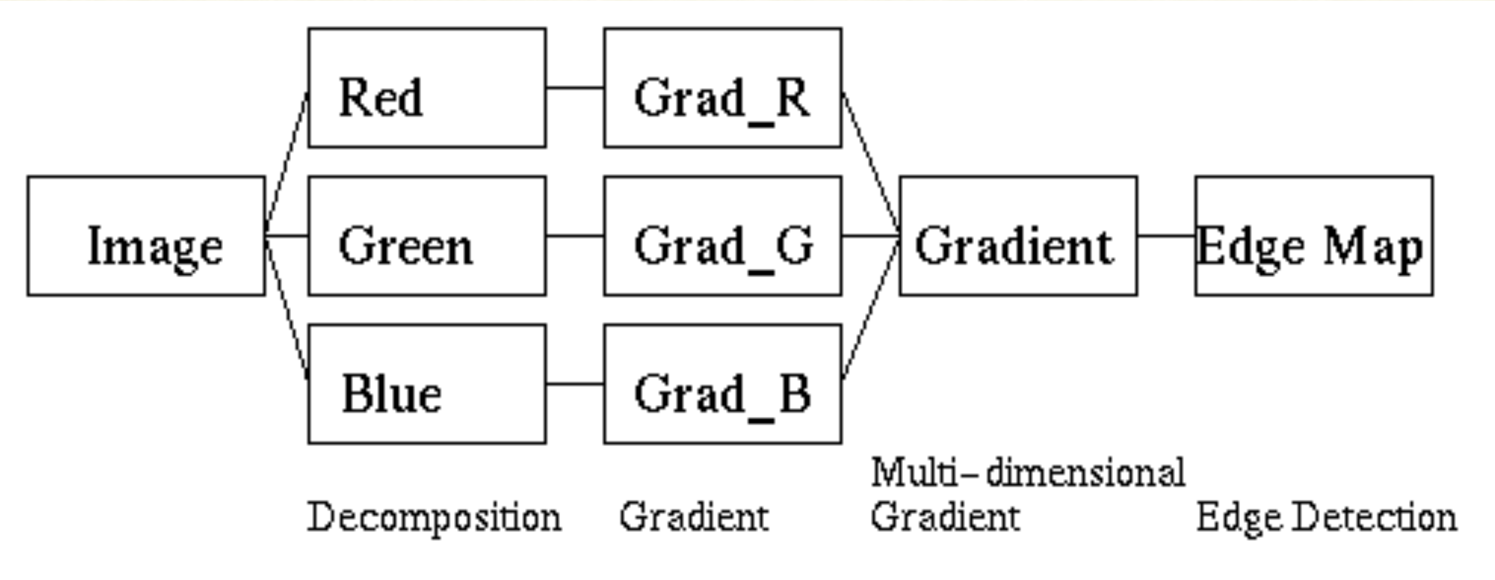
Color Image Filtering

- Median filtering



Color Edge Detection





Edge in Color Images

References

- “Colorimetry”, Ohta and Robertson, John Wiley and Sons Ltd
- https://hypjudy.github.io/2017/03/19/dip-histogram-equalization/
- http://www.inf.u-szeged.hu/ssip/2008/presentations2/Gordan_LectureSSIP08.pdf

Questions?